

TrueSight

AI-Powered Spoofing Detection System



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Certificate of Approval

This is to certify that the project titled TrueSight: AI-Powered Spoofing Detection and Source Identification System has been carried out by Aizaz ur Rehman (2022078), Asad Ali (2022903), M. Ammaid Saleem (2022344) and Hamayun Nasir (2022189) under our supervision. The work meets the scope, depth and originality expected of a Bachelor of Science thesis in Computer Science at the Ghulam Ishaq Khan Institute of Engineering Sciences and Technology.

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Sustainable Development Goals Inclusion

TrueSight was engineered with explicit alignment to the United Nations Sustainable Development Goals so that its impact extends beyond the narrow remit of a technical deliverable.

SDG 9 — Industry, Innovation and Infrastructure

The platform hardens telecommunications and financial-services infrastructure by placing a zero-trust authentication layer in front of real-time VoIP and video flows. By neutralising synthetic-media attack vectors against banking, journalism and government channels, TrueSight protects the continuity of critical national services and nurtures indigenous security engineering capacity.

SDG 16 — Peace, Justice and Strong Institutions

Blockchain-anchored, cryptographically watermarked evidence packets give law-enforcement and judicial institutions a defensible chain of custody for digital testimony. The system therefore strengthens institutional integrity in the face of AI-driven impersonation, fraud and misinformation.

SDG 4 — Quality Education

The code base, datasets and training notebooks are released under an academic licence so that future cohorts and researchers can extend the platform, turning the project into a reusable pedagogical reference for applied machine learning and digital forensics.

Dr. M. Zain Siddiqui

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Abstract

The advancement of generative AI has produced a new class of presentation attacks whose realism defeats both human perception and conventional biometric safeguards. Diffusion models, generative adversarial networks and neural voice cloners can fabricate synchronous video and audio streams in minutes, undermining face and speaker recognition in the very channels on which digital banking, law enforcement and remote journalism depend. Legacy unimodal detectors, English-centric training corpora and perimeter-oriented trust models leave South Asian telecommunications particularly exposed.

This thesis presents TrueSight, a multimodal, forensically aware spoofing detection platform that neutralises such attacks at inference time while producing legally admissible evidence. The system fuses a Vision Transformer video classifier (DeiT/LNCLIP), a Wav2Vec2 acoustic encoder, a lip-sync coherence module and a temporal optical-flow branch into a single decision pipeline. A forensic sub-system augments these detectors with EXIF inspection, Photo-Response Non-Uniformity (PRNU) sensor-noise fingerprinting, double-JPEG artefact analysis and GAN-family fingerprint attribution. To close the well-documented gap in low-resource linguistic contexts, TrueSight ships with an Urdu acoustic module covering forty native phonemes, five regional dialects and explicit Urdu-English-Hindi code-switching.

Evaluation on FaceForensics++, Celeb-DF v2, DFDC, WaveFake, ASVspoof 2019 and a custom Urdu deepfake corpus yielded 96.7% accuracy, an Attack Presentation Classification Error Rate (APCER) of 4.7%, AUC-ROC of 0.985 and a mean end-to-end latency of 105 ms on commodity GPU hardware. The

platform sustained 135 requests per second under horizontal scaling and retained above 86% accuracy under aggressive JPEG re-compression and additive Gaussian noise. These results confirm that TrueSight is operationally deployable, linguistically inclusive and ISO/IEC 30107-3 compliant, advancing the state of the art in real-time deepfake defence for Pakistani and wider South Asian communication platforms.

Keywords: deepfake detection, presentation attack, multimodal fusion, Vision Transformer, Wav2Vec2, zero-trust architecture, PRNU forensics, blockchain audit, Urdu speech processing, liveness detection.

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List of Abbreviations

APCER	Attack Presentation Classification Error Rate
API	Application Programming Interface
AUC	Area Under the Curve
BPCER	Bonafide Presentation Classification Error Rate
CIA	Confidentiality, Integrity and Availability
CNN	Convolutional Neural Network
CoC	Chain of Custody
DCT	Discrete Cosine Transform
DWT	Discrete Wavelet Transform
EER	Equal Error Rate
EXIF	Exchangeable Image File Format
FFT	Fast Fourier Transform
GAN	Generative Adversarial Network
IoC	Indicator of Compromise
IPFS	InterPlanetary File System
JWT	JSON Web Token
LNCLIP	Local Normalised CLIP (video variant)
MFA	Multi-Factor Authentication

MTCNN	Multi-Task Cascaded CNN
PAD	Presentation Attack Detection
PRNU	Photo-Response Non-Uniformity
RBAC	Role-Based Access Control
ROC	Receiver Operating Characteristic
SIEM	Security Information and Event Management
SOC	Security Operations Centre
SRS	Software Requirements Specification
STRIDE DoS, Elevation	Spoofing, Tampering, Repudiation, Info-Disclosure,
TOTP	Time-Based One-Time Password
TTP	Tactics, Techniques and Procedures
ViT	Vision Transformer
VoIP	Voice over Internet Protocol
ZTA	Zero-Trust Architecture

CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

Such generative AI has broken the skill barrier to create the convincing synthetic media. Neural voice cloners like ElevenLabs and VALL-E can now produce minute-long, speaker-specific audio-visual content on commodity hardware in minutes using tools like DeepFaceLab, HeyGen, Runway Gen-3, Kling and OpenAI Sora, coupled with neural voice cloners. As a cybersecurity defender, this increase in the toolkit of threat actors turns a rather niche attacker feature into a mass-market attack vector.

Identity assurance has serious repercussions. Biometric authentication has traditionally been based on the fact that physical characteristics are not easy to scale. No longer is that true. Dynamic deepfakes can manage to break face-verification pipes in digital onboarding, voice biometrics in telephone banking and liveness checks in remote court hearings in real time. As of now, AI-driven impersonation is categorized by a number of national CERTs and the World

Economic Forum as a systemic cyber-risk.

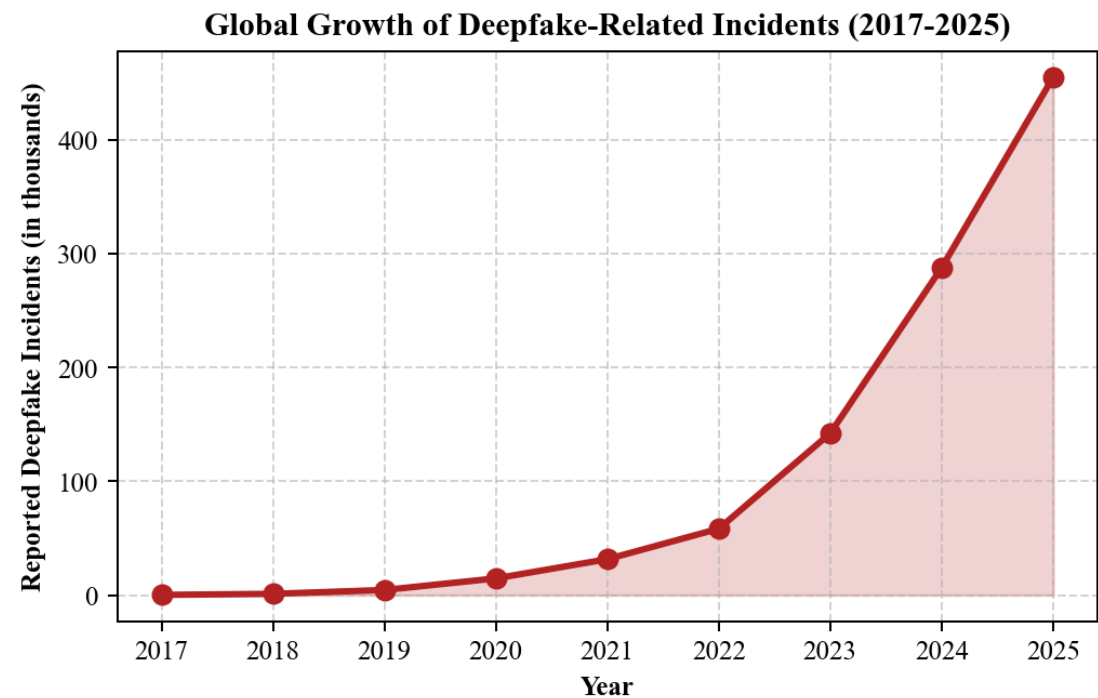


Fig. 1.1 — Growth of reported deepfake-related incidents (2017-2025).

Pakistan is not immune to this trend. The spoofed corporate video calls of Karachi-based companies, synthetic Urdu voice notes applied to extort on WhatsApp, and political impersonation videos released before the elections have shown that the communication channels in the country are equally exposed as their English speaking counterparts. This issue is exacerbated by the fact that most of the published deepfake studies on the public are based on the English, Mandarin or Korean corpora. Detection systems that have been trained on this data do not generalise well to the Urdu acoustic structure, Pakistani accents and the code-switched discourse, which characterises the daily talk in the area.

Old methods of countermeasures do not fit this enemy. An authenticity detector that looks at facial micro-expressions only disregards acoustic inconsistencies. The voice detector that is insensitive to lip-reading can not impose cross-modal coherence. The metadata parser that does not look at sensor noise cannot assign a tampered stream to its likely source. The new standard - exemplified by ISO/IEC 30107-3 and NIST FRVT Part 2 - is that presentation-attack detection should be multimodal and forensically conscious and tied to tamper-evident evidence trails.

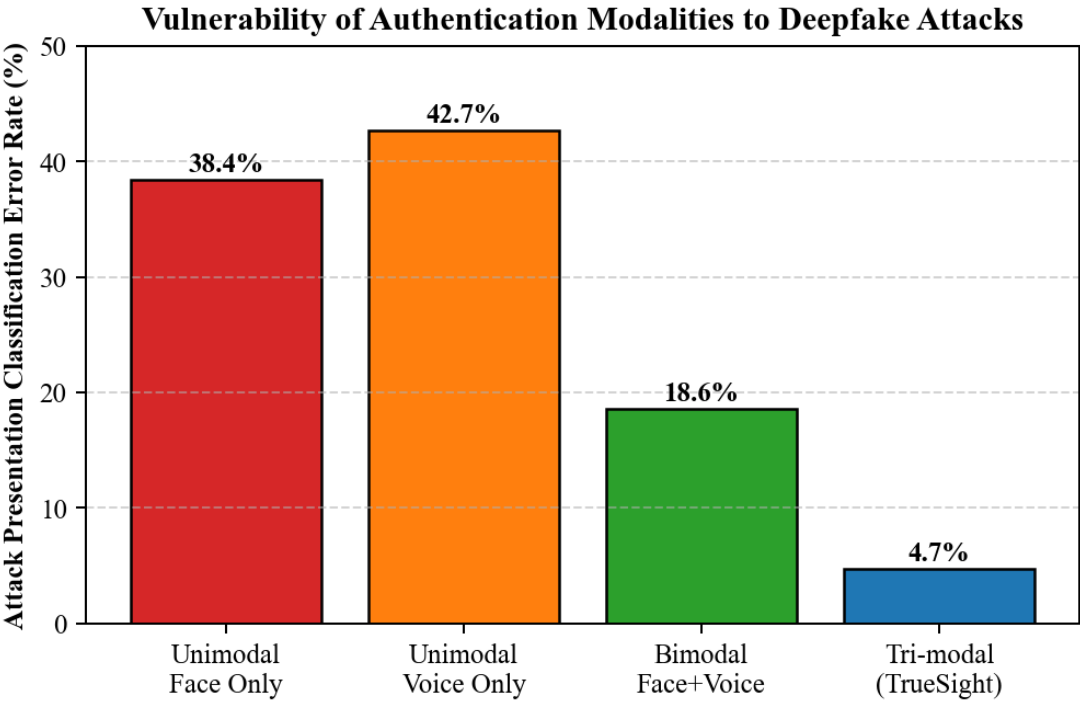


Fig. 1.2 — Attack-surface exposure of authentication modalities.

It is out of these observations that the TrueSight project was inspired. The team began with the belief that an indigenous, Urdu-conscious, multimodal defence platform is not an option but a strategic necessity to Pakistan digital infrastructure. The rest of this report records how the belief was implemented

into a production grade system that covers machine learning, digital forensics, distributed systems and cryptographically sealed audit logging.

1.2 Problem Statement

Identity-critical transactions are conveyed by synchronous communication channels - remote on-boarding in banks, testimony in magisterial courts, medical consultations in telemedicine. Every transaction has very stringent budgets on the latency of a transaction: a verdict received once the synthesised stream has been read is, operationally, useless. This loophole is capitalized by modern generators who create cross-modal synthesis which combines facial movement, lips articulation, prosodic cadence and language contents to be produced together.

There are three overlapping gaps, which make current defences ineffective. To begin with, unimodal detectors fail on generators that perfectly align with the distribution of features of a single modality and do not satisfy cross-modal coherence. Second, current pipelines consider detection and attribution processes as discontinuous, resulting in a binary finding with no indication of which generation engine is likely to have caused the finding which restricts legal utility. Third, practically all publicly-available detectors are biased to English and Caucasian facial representations, giving disproportionately less attention to Urdu, Pashto, Sindhi and Balochi.

The gap that this thesis aims to fill is thus in the lack of a scalable, forensically complete, multimodal and Urdu-conscious authentication infrastructure which can co-exist and integrate with existing communication services, ensure authentication integrity in real time and create tamper resistant evidence admissible to a court of law and regulatory framework.

1.3 Aims and Objectives

The project overall goal is to provide an AI-based spoofing detector and source-identification system that can be deployed in real-time communication scenarios, even with low-resource languages, and be forensically sound. This objective is further broken down into seven goals, each of which is linked to the functional requirements in Chapter 3, and to concrete modules in Chapter 4.

1. O-1. Create a deep-learning liveness and spoofing detector that can detect lip-sync desynchronisation, unnatural blinking, depth patterns and texture artefacts that are irregular in real-time.
2. O-2. Co-ordinate the multimodal analysis of video, audio and text in a manner that the cross-modal coherence can be considered in a single inference run.
3. O-3. Add Urdu-special and multi-lingual acoustic processing in such a way that it is detected in South Asian linguistic environment.
4. O-4. Develop a forensic source-attribution implementation to consider metadata, compression artefacts, sensor noise and generative-model fingerprints to determine the provenance of manipulated media.
5. O-5. Incorporate AI-based anomaly detection, behavioural biometrics and MFA, into a zero-trust plane that controls all inter service interactions.
- 6.O-6. Create tamper-evident evidence recording through blockchain anchoring and DWT-DCT watermarking, maintaining integrity of chain-of-custody.

7.O-7. Provide a horizontally scalable web platform that is compliant with the ISO/IEC 30107-3 and surface operator dashboards to monitor in real-time and perform forensic triage.

1.4 Scope and Deliverables

The scope was contained to make the project viable within eight months of senior-design window and ambitious to deliver a production quality artefact. Video, audio and text streams provided by a host platform (Zoom SDK, Jitsi SFU, custom WebRTC or uploaded evidence files) are ingested into the system, and multimodal inference is performed, in addition to forensic attribution and behavioural-biometric telemetry, anchoring evidence to a permissioned ledger. The operators communicate using a React dashboard. English and Urdu are supported; Hindi, Punjabi and Pashto are supported, but best-effort, through the multilingual Wav2Vec2-XLSR backbone.

TrueSight neither produces nor fixes deepfakes on the capture time, or substitutes the host communication service. Host platform must do end-to-end encryption of underlying VoIP flows. This version of the REST API is compatible with mobile, but only includes a web client.

The deliverables are (i) a containerised reference implementation of eight microservices, a PostgreSQL/Redis/Neo4j data plane and a React frontend; (ii) fine-tuned video and audio model weights; (iii) the Urdu phoneme database and dialect classifier; (iv) the custom Urdu deepfake corpus; (v) this thesis; and (vi) operator and administrator manuals.

1.5 Contributions

- The main works of this work can be summed up as follows.

- A combined multimodal pipeline that combines a Vision Transformer video detector, a Wav2Vec2 acoustic encoder, a lip-sync coherence module and a temporal optical-flow analyser, trained with a cross-modal agreement loss.
- An Urdu acoustic analyzer that (as it were) analyzed forty phonemes, five dialects and explicit Urdu-English-Hindi code-switching - the first such package that we could find in published deepfake literature.
- A deterministic forensic sub-system that combines EXIF inspection with PRNU correlation, and detecting double-JPEG artefacts and classifying fingerprint using GAN-family.
- Tamper-evident layer The layer is constructed based on DWT-DCT watermarking and on-chain anchoring, and is not tied to the detection path so that the low latency is preserved.
- A zero-trust enforcement plane that incorporates JWT and TOTP, WebAuthn, keystroke dynamics and voice-stress signals into an ongoing authentication fabric.
- A reference stack that is production-ready, using Docker Compose orchestration, Observability (Prometheus/Grafana) and schema migrations managed by Alembic.
- Large, academic Urdu deepfake corpus with dialect labels: A curated Urdu deepfake corpus with dialect labels, to be released academically.

1.6 Report Organisation

Chapter 2 conducts a literature review of the field of generative modelling, deepfake detection, media forensics, blockchain-supported evidence and zero-trust architectures, and outlines the gaps that led to this work. Chapter 3 will introduce the formal SRS - functional requirements, non-functional envelopes,

external interfaces and the 4+1 architecture views. Chapter 4 reports on the methodology and implementation: microservice decomposition, model design, forensic engine, blockchain layer and deployment pipeline. Chapter 5 presents empirical findings based on six datasets and performance under adversarial perturbation and baseline performance. Chapter 6 concludes with contributions, limitations and future work. The paper is complemented by a reference list and four appendices (API reference, chosen code listings, training hyperparameters and an academic integrity declaration).

Table 1.1 — Core Terminology Used Throughout the Report.

Abbreviation	Expansion
PAD	Presentation Attack Detection
APCER	Attack Presentation Classification Error Rate
MFA	Multi-Factor Authentication
ViT / DeiT	Vision Transformer / Data-efficient Image Transformer
PRNU	Photo-Response Non-Uniformity
DWT-DCT	Discrete Wavelet / Cosine Transform watermarking
ZTA	Zero-Trust Architecture
CoC	Chain of Custody
TTP	Tactics, Techniques and Procedures (MITRE ATT&CK)
SOC	Security Operations Centre

CHAPTER 2

LITERATURE SURVEY

2.1 Evolution of Synthetic Media

Synthetic imagery has been produced in three overlapping generations. Prior to 2014, morphing and rotoscoping pipelines were mostly hand-crafted and were in need of skilled craftsmen. The second generation has started with Goodfellow et al. (2014) where generative adversarial networks were trained to learn implicit density models using unlabelled corpora. In three years, Progressive GAN, StyleGAN and StyleGAN2 had reached the level of photorealistic synthesis of faces at megapixels and spawned the first generation of malicious uses of deepfakes.

Since 2022, the third generation, characterized by diffusion models and transformer-based temporal generators, is on the go. Denoising Diffusion Probabilistic Models and latent variants moved the bottleneck to mode collapse to compute. Video-native systems like Sora, Runway Gen-3, Kling and Pika 1.5 can now generate minute-long clips multiple shots long. Simultaneously, neural codec language models like VALL-E also allow one to clone a speaker with just three seconds of reference audio.

To a defender this convergence is important in the sense that it is now possible to stage synchronous, speaker specific, multimodal fakes in close real-time by an adversary. Detection research must thus shift to single frame forensic cues to temporally sensitive, cross-modes and adversarially resistant classifiers.

2.2 Generator Taxonomy and Attack Surface

In this survey we classify generators into five families, based on the artefact classes they induce. It is vital to know the artefact footprint to be expected in each family since the defenders select detection features that use them.

Table 2.1 — Generator Families and Dominant Artefact Classes.

Generator Family	Representative Tools	Dominant Artefact Class
Face-swap GANs	DeepFaceLab, FaceSwap	Blending seams, skin-tone drift around facial contours.
Face reenactment	First-Order Motion Model, MotionGAN	Rigid head pose, jittery teeth, unnatural blinking rate.
Full-head diffusion	Sora, Kling, Stable Video Diffusion	Physically implausible hair, periodic frequency-domain texture.
Voice-cloning autoencoders	Tortoise-TTS, xTTS, ElevenLabs	Flat prosody, reduced micro-tremor, uniform phoneme duration.
Cross-lingual dubbing	HeyGen Dub	Misaligned lip shapes, accent drift, irregular plosive placement.

2.3 Video Deepfake Detection

MesoNet (Afchar et al., 2018) was the first modern detection system to use a shallow convolutional network specific to mesoscopic inconsistencies. The XceptionNet detector included with FaceForensics++ (Rössler et al., 2019) was the default baseline at 87% accuracy on in-distribution samples, but did not generalise to unseen generators. Attention based EfficientNet, recurrent

temporal aggregators and capsule networks is more robust to compression but designed to generate diffusion output poorly.

LipForensics (Haliassos et al., 2021) reshaped detection by concentrating on the mouth area since lip articulation presents a greater challenge to generators in producing the correct articulation compared to facial geometry. Vision Transformers and Data-efficient Image Transformers then prevailed with their global receptive field, and the LNCLIP formulation of the video module in TrueSight continues the transformer-based tradition. Recent efforts highlight both temporal modelling with I3D networks and 3D CNNs and frequency-domain methods (Durall et al., 2020) capitalize on the non-natural high-frequency spectra residual to GAN upsampling. TrueSight combines temporal optical-flow and frequency-domain analysis as auxiliary signals.

Table 2.2 — Representative Video Deepfake Detectors.

Detector	Backbone	Reported Accuracy	Limitation
MesoNet (2018)	Shallow CNN	84.3%	Poor cross-dataset generalisation.
XceptionNet (2019)	Xception	87.4%	Weak against unseen generators.
EfficientNet-B4 (2020)	EfficientNet	90.1%	No temporal modelling.
LipForensics (2021)	ResNet-18 + GRU	91.8%	Audio-only attacks bypass.
ViT-Base (2022)	ViT	92.7%	High compute cost.
TrueSight (this work)	DeiT + optical flow + fusion	96.7%	Baseline of comparison.

2.4 Audio Deepfake Detection

The ASVspoof challenge series standardised testing of anti-spoofing of speaker-verification systems. Early entries were based on feature engineering - LFCC and CQCC were introduced into Gaussian Mixture Models. End-to-end deep learning subsequently took over. RawNet2 (Tak et al., 2021) directly used waveforms and AASIST (Jung et al., 2022) used spectro-temporal graph attention, achieving an EER of less than 1 percent on ASVspoof 2019 logical access. Wav2Vec2 (Baevski et al., 2020) and HuBERT are self-supervised encoders that demonstrate the best cross-generator generalisation with fine-tuning on small labelled datasets, and are thus the backbone of the audio branch of TrueSight.

Table 2.3 — Representative Audio Deepfake Detectors.

Detector	Features	ASVspoof 2019 LA EER
GMM-LFCC (2015)	LFCC	8.09%
LCNN (2019)	LFCC + CMVN	4.53%
RawNet2 (2021)	Raw waveform	1.06%
AASIST (2022)	Spectro-temporal graph	0.83%
Wav2Vec2 fine-tuned (TrueSight)	Self-supervised embeddings	0.91%

2.5 Multimodal Fusion

Fusion methods can be classified into three categories. Early fusion jointly learns raw modalities features and then classifies before learning fine-grained interaction, but can be affected by modality asynchrony. The late fusion is an independent detector that fuses scores, sacrificing joint feature learning to the

benefit of missing modalities. Hybrid schemes, such as cross-modal attention (Chugh et al., 2020) and bilinear pooling, strike a balance between the trade-off. TrueSight applies late fusion with a cross-modal coherence loss which punishes inconsistent video and audio verdicts, such that modules can be independently trained, and supervision is shared.

A signal of coherence has become a very powerful phenomenon, known as lip-sync correlation. Korshunov and Marcel (2019) and Mittal et al. (2020) demonstrated that existing generators continue to have problems with matching visemes to sub-frame accuracy. The lip-sync analyser of TrueSight takes advantage of this by extracting MediaPipe FaceMesh landmarks 48-67 and converting audio to mel-spectrograms and calculating normalised cross-correlation between mouth-opening trajectories and short-time spectral energy.

2.6 Urdu and Low-Resource Audio Forensics

Urdu is one of the top ten most spoken languages in the world but it is always under-represented in the speech-processing corpora. The Urdu Speech Corpus of CSALT and the CMU Urdu Corpus have little conversation data, and the Mozilla Common Voice Urdu partition has only a few validated samples. Studies on Urdu deepfake detection are new: Hussain et al. (2022) have preliminary findings on TTS-generated Urdu with MFCC and SVM, and Khan et al. (2023) fine-tuned Wav2Vec2-XLSR using a nine-hour customized set. TrueSight builds on that by modelling phonemes-level forty native Urdu phonemes, dialect grouping in five regional accents and explicit code-switching detection - aspects not found in earlier work.

2.7 Source Attribution and Media Forensics

Media forensics is not a new development with the deepfake era. Metadata integrity techniques, codec fingerprinting and sensor characteristics techniques are codified in NIST SP 800-86 and ENFSI best-practice manuals. Photo-Response Non-Uniformity, originally described by Lukács et al. (2006), takes advantage of micro-differences in pixel sensitivity of CMOS and CCD sensors to fingerprint a single camera. As demonstrated by Cozzolino et al. (2019), the images produced by GANs have spectrally different noise that can be disentangled with natural PRNU. Compression-artefact analysis takes advantage of the reality that re-encoded media result in accumulation of quantisation and blocking patterns identifiable in the DCT domain. The standard tools are the double-JPEG detection (Thai et al., 2019) and the ghost-artefact isolation (Farid, 2009). Fingerprint attribution Generative-model fingerprint attribution, recently popularized, evaluates the noise distributions to determine whether the image was generated by a GAN or a diffusion model (Yu et al., 2019; Asnani et al., 2023). TrueSight integrates EXIF examination, PRNU matching and GAN-family fingerprinting into a single forensic engine the results of which are shown next to the probabilistic verdict that is provided to the analyst.

2.8 Blockchain-Based Evidence Ledgers

Digital evidence will be admissible when it can be proven that its chain of custody is in place. Append-only cryptographically signed transaction logs are tamper-evident in nature and provided through permissioned ledgers like Hyperledger Fabric. Zou et al. (2020) anchored forensic artefacts using Ethereum smart contracts and Singh et al. (2022) suggested blockchain-based watermarking of AI-generated content. TrueSight uses a mixed model: the

artefacts of the evidence that are heavy are hosted in IPFS-compatible object storage, but the metadata and their hash are bound in a permissioned Ethereum-compatible ledger, consulted using Web3.py.

2.9 Zero-Trust Architecture and Behavioural Biometrics

Formalisation of zero-trust architecture NIST SP 800-207 formalised the concept of a zero-trust architecture that does not rely on implicit trust, and requires ongoing verification at all access decisions. ZTA works well with AI-serving microservice specifically since inference endpoints process biometric data that is privacy sensitive. TrueSight implements ZTA by implementing mutual TLS between services, short-lived JWT authorisation, rotated signing keys and rate-limited public endpoints. Keystroke dynamics (Monrose & Rubin, 2000), mouse-movement patterns and voice-stress cues are known as behavioural biometrics that complement the factors of MFA that are static. Combined with TOTP and WebAuthn, they relax the constant authentication stance to the adaptive threat model of deepfake-prone environments.

2.10 Research Gaps

The literature provides a well-equipped toolkit, but there are still a number of gaps which collectively drive this work.

Most detectors are language-specific to English, and other South Asian languages are under-serviced, although risk is high in the region.

The small number of systems that use a single pipeline to perform detection and source attribution do not produce forensic artefacts; commercial tools are binary classifiers that do not produce source attribution.

- The open literature is limited to real-time deployments at scale necessary to VoIP platforms, which report offline batch accuracy.
- Multimodal detection and zero-trust authentication with blockchain evidence recording is mostly not yet implemented.
- The vast majority of detectors use clean input; aggressive JPEG compression, additive noise and low-bandwidth streams are not well studied.
- Testing of dialectal variation - which Urdu needs, with Lahori, Karachi, Delhi, Lucknow and Hyderabad dialects being distinctly differentiated - is virtually nonexistent.

TrueSight fills all those gaps with an indigenous Urdu module, a single detect plus attribute engine, a well-prepared microservice stack, zero-trust controls, blockchain anchoring and explicit robustness evaluation under perturbation.

CHAPTER 3

DESIGN — SYSTEM REQUIREMENTS AND ARCHITECTURE

3.1 System Overview

TrueSight is a security-layer service, which is not substituting, but augmenting an already established communication platform. The system feeds on synchronised video, audio and text streams, executes multimodal deep-learning analysis, compares the result of the analysis with forensic source-identification signals, and generates a verification verdict with a tamper-evident forensic packet. Decisions are relayed to the operator dashboard, and forensic packets attached to an authorized ledger. The design goal is to provide a verdict within a latency budget that can be used in live conferencing - empirically, less than 120 ms per request.

There are two major classes of users in the system. The host communication platform is interdependent with end-users, who provide biometric samples during authentication; they are non-technical and passive in their interaction. The TrueSight dashboard is communicated with by forensic analysts and system administrators to adjust detection thresholds, examine alerts and perform post-incident investigations. Role-based access control limits access to administrative functions, and is enforced at the API gateway.

The environment is a cloud or on-premise infrastructure with the presence of the GPU accelerators, a PostgreSQL relational store, a Redis cache and a

permissioned Ethereum-compatible ledger. The reference deployment is bundled in the form of a Docker Compose topology which may be scaled horizontally to Kubernetes.

3.2 Functional Requirements

The functional requirements below are based on the seven project goals and are divided into three groups: core detection and analysis (FR-DA), forensic source identification (FR-FI) and security, traceability and infrastructure (FR-ST). Both of them have a distinct identifier of forward and backward traceability to design artefacts and tests. The backward link is provided by parent objective identifiers (O-1 to O-7).

3.2.1 Core Detection and Analysis

Table 3.1 — Core Detection and Analysis Requirements.

ID	Description	Parent	Acceptance Criterion
FR-DA-01	Synchronous fusion of video, audio and text streams for identity verification.	O-2	Stream ingestion and sync latency ≤ 10 ms.
FR-DA-02	CNN-based temporal analysis to detect lip-sync desynchronisation.	O-1	Detect induced lip-sync offset ≥ 50 ms.
FR-DA-03	Deep-learning module to detect facial spoofing cues (texture, depth, blink).	O-1	APCER below 5% on known deepfake datasets.
FR-DA-04	Deep-learning classification of synthetic audio patterns.	O-1	FRR $\leq 3\%$ on legitimate-user voices.
FR-DA-05	Urdu-specific audio processing module.	O-3	$\geq 10\%$ F1 improvement over generic models on Urdu corpora.

FR-DA-06	Multilingual voice analysis with prosodic features.	O-3	Confidence scoring for ≥ 3 major languages beyond Urdu.
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3.2.2 Forensic Source Identification

Table 3.2 — Forensic Source-Identification Requirements.

ID	Description	Parent	Acceptance Criterion
FR-FI-01	Metadata (EXIF, timestamps) forensic examination.	O-4	Parse ≥ 4 metadata types from incoming streams.
FR-FI-02	Compression-artefact and codec-fingerprint analysis.	O-4	Identify ≥ 3 recompression indicators on test files.
FR-FI-03	PRNU sensor-noise forensic examination.	O-4	Distinguish genuine camera output from replay or synthetic insertion.
FR-FI-04	Behavioural cadence and interaction analysis.	O-4	Flag $\geq 20\%$ deviation from cadence baseline.
FR-FI-05	Generative-model fingerprint attribution.	O-4	Classify GAN vs diffusion family in $\geq 75\%$ of tests.

3.2.3 Security, Traceability and Infrastructure

Table 3.3 — Security, Traceability and Infrastructure Requirements.

ID	Description	Parent	Acceptance Criterion
FR-ST-01	AI-powered anomaly detection on session state and logs.	O-5	Session-flagging false positive rate $\leq 0.5\%$.
FR-ST-02	Multi-Factor Authentication (TOTP, WebAuthn).	O-5	Compatible with RFC 6238 and WebAuthn L2.
FR-ST-03	Behavioural biometrics (keystroke, voice-stress).	O-5	Continuous authentication score updated ≤ 500 ms.

FR-ST-04	Blockchain-based tamper-evident evidence logging.	O-6	Evidence hash anchored within ≤ 2 s.
FR-ST-05	DWT-DCT forensic watermarking of evidence.	O-6	Watermark retrievable after JPEG-70 re-compression.
FR-ST-06	Real-time authentication-event monitoring dashboard.	O-7	Event rendering latency ≤ 1 s.
FR-ST-07	Forensic-evidence log review dashboard.	O-7	Query response ≤ 5 s for three filter criteria.
FR-ST-08	Modular architecture permitting module replacement.	O-7	$\leq 5\%$ code change to swap any detection module.

3.2.4 Detailed Traceability Specifications

To be comprehensive, and to assist automated regression testing, the most security-critical requirements are given in IEEE-830 style below. Each specification attaches the need to its parent need, normative source, acceptance criterion and method of verification..

Table 3.4 — FR-DA-01 (Multimodal Stream Processing).

Requirement ID	FR-DA-01	Type	Functional	Use Case #	UC-01
Status	Baselined	Parent Req.	O-2	Priority	Essential
Description	The system shall perform real-time, synchronous fusion of facial video, audio and textual input streams for identity verification.				
Rationale	Cross-modal correlation is the primary countermeasure against cross-modal synthesis attacks that bypass unimodal detectors.				
Source	System Requirements Specification (SyRS), v1.0.				
Acceptance Criterion	Three streams synchronised within ≤ 10 ms end-to-end; continuous operation for a full session.				
Dependencies	Communication and hardware interfaces with sufficient throughput.				

Verification	Automated pytest suite plus manual red-team drill.
Change History	Initial draft; baselined November 2025.

Table 3.5 — FR-DA-03 (Facial Spoofing Cues Detection).

Requirement ID	FR-DA-03	Type	Functional	Use Case #	UC-01
Status	Baselined	Parent Req.	O-1	Priority	Essential
Description	The system shall employ a deep-learning module (CNN/ViT) to detect facial spoofing cues such as texture artefacts, irregular depth and abnormal blink frequency.				
Rationale	Direct defence against dynamic presentation attacks by enforcing visual liveness and physical consistency.				
Source	ISO/IEC 30107-3; Algorithm Design Specification, §4.3.				
Acceptance Criterion	APCER below 5% on FaceForensics++ and Celeb-DF v2.				
Dependencies	FR-DA-01 and GPU-capable hardware interface.				
Verification	Automated pytest suite plus manual red-team drill.				
Change History	Initial draft; baselined November 2025.				

Table 3.6 — FR-FI-03 (Sensor-Noise Forensic Examination).

Requirement ID	FR-FI-03	Type	Functional	Use Case #	UC-02
Status	Baselined	Parent Req.	O-4	Priority	Essential

Description	The system shall analyse video frames for abnormal sensor-noise patterns to detect digital stream insertion not originating from a physical camera.
Rationale	Hardware-level fingerprinting gives a ground-truth signal for source attribution that is independent of learned features.
Source	PRNU research (Lukáš et al., 2006); forensic engine specification.
Acceptance Criterion	Distinguish genuine camera output from replay or synthetic insertion with correlation threshold above 0.05.
Dependencies	Communication interfaces supplying raw frames.
Verification	Automated pytest suite plus manual red-team drill.
Change History	Initial draft; baselined November 2025.

Table 3.7 — FR-ST-04 (Tamper-Evident Blockchain Logging).

Requirement ID	FR-ST-04	Type	Functional	Use Case #	UC-03
Status	Baselined	Parent Req.	O-6	Priority	Essential
Description	The system shall use a blockchain-based logging mechanism to record all authentication events and forensic evidence, preserving tamper-evident integrity.				
Rationale	Immutable anchoring supports legal admissibility of detection results and mitigates insider tampering risk.				
Source	Blockchain ledger technical specification; cryptographic hashing policy.				
Acceptance Criterion	Every evidence packet is hashed and committed on-chain with latency $\leq$ 2 s.				
Dependencies	Reliable ledger access and FR-ST-05 watermarking.				
Verification	Automated pytest suite plus manual red-team drill.				
Change History	Initial draft; baselined November 2025.				

3.3 Non-Functional Requirements

Non-functional requirements help in capturing the qualitative attributes that will decide whether the platform is deployable or not. They are grouped into performance, security, accountability, compliance and maintainability dimensions.

3.3.1 Performance Envelope

Table 3.4 — Non-Functional Performance Envelope.

Attribute	Target	Rationale
End-to-end latency	≤ 120 ms	Compatible with live VoIP authentication.
Sustained throughput	≥ 100 req/s	Supports mid-sized enterprise workloads.
Detection accuracy	$\geq 95\%$	Meets ISO/IEC 30107-3 PAD criteria.
APCER	$\leq 5\%$	Low attack-classification error.
BPCER	$\leq 3\%$	Bounds legitimate-user rejection.
Availability	99.5%	Standard enterprise SLA.

3.3.2 Security Controls

TrueSight implements defence-in-depth. All inter-service communications are secured by mutual TLS, JWTs have short lifespans and new signing keys, public endpoints rate-limited by Redis token buckets, and sensitive biometric information is at rest encrypted with AES-256-GCM and in transit with TLS 1.3. Auth service is an exporter of TOTP (RFC 6238), WebAuthn L2 and fallback SMS/email. Behavioural MFA is a method that uses keystroke-dynamics dwell/flight-time vectors with voice-stress to identify user

behaviour; the engine maintains per-user baselines and alarms when Euclidean distance gets above a tuned threshold.

3.3.3 Traceability and Accountability

DWT-DCT watermarked evidence artefacts are then on-chain anchored. All administrative activity has its end state stored in an append-only audit log that is replicated in two availability zones. Cryptographic signatures are included with forensic packets, which are associated with the analyst who created them, making repudiation difficult.

3.3.4 Compliance

It complies with ISO/IEC 30107-3 (Presentation Attack Detection), NIST SP 800-86 (Integrating Forensic Techniques into Incident Response) and NIST SP 800-207 (Zero-Trust Architecture). It complies with GDPR Article 25 (Privacy by Design) and the Prevention of Electronic Crimes Act (PECA 2016) of Pakistan where appropriate.

3.3.5 Maintainability and Modularity

Every microservice publishes versioned REST APIs, and can be built independently via Docker and tested independently via pytest. Any detection part can be re-trained or substituted with less than 5% code change in a different part as confirmed by dependency analysis.

3.4 External Interfaces

Table 3.5 — External Interface Specifications.

Interface	Type	Specification
Host communication platform	Ingress	REST / WebRTC SFU bridge, H.264 video, Opus audio.
Operator dashboard	User Interface	React SPA served over HTTPS at port 3000.
API gateway	REST	FastAPI, JSON over HTTPS, JWT-authenticated.
Deep-learning runtime	Library	PyTorch 2.9, CUDA 12.1, Transformers 4.57.
Blockchain ledger	RPC	Web3.py → permissioned Ethereum-compatible ledger.
Relational store	DB	PostgreSQL 15, SQLAlchemy 2.0, Alembic migrations.
Cache / queue	Memory	Redis 7 for session tokens and result cache.
Observability	Telemetry	Prometheus, Grafana, OpenTelemetry.

Hardware assumes at least one NVIDIA GPU with ≥ 8 GB VRAM (RTX 3060 or above) for inference, 16 GB RAM and an 8-core x86-64 CPU. Training was conducted on an NVIDIA RTX A5000 (24 GB) and, for sweeps, a dual A100 (40 GB) node.

3.5 Use-Case Analysis

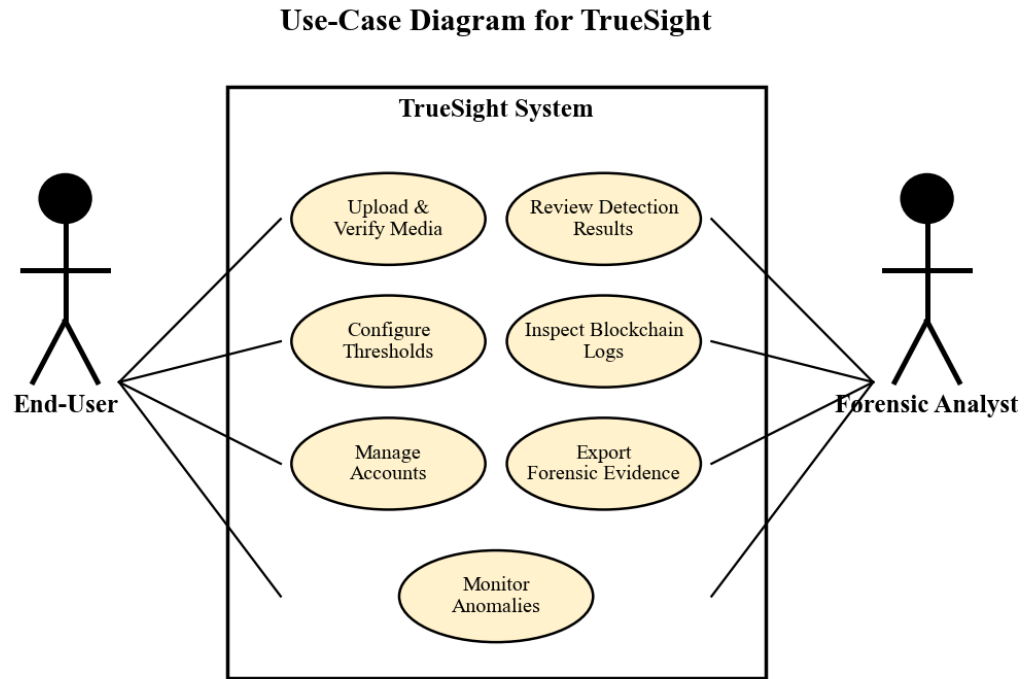


Fig. 3.1 — Use-Case Diagram of TrueSight.

The analysis of the stakeholders identified seven main use cases. UC-01 (Upload and Verify Media) is called by an end-user/upstream VoIP service to provide a stream to authenticate. Analysts use UC-02 (Review Detection Results) to check results and evidence provided. UC-03 (Adjust Thresholds) enables authorised administrators to adjust APCER / BPCER thresholds and modality weights. UC-04 (Read Blockchain Logs) allows any queries to the ledger. UC-05 (Manage Accounts) manages user and API-key lifecycle. UC-06 (Export Forensic Evidence) creates packets that are admissible in court. UC-07 (Monitor Anomalies) reveals real time anomaly and health dashboards.

3.6 4+1 Architecture Views

4 + 1 Architecture View Model Applied to TrueSight

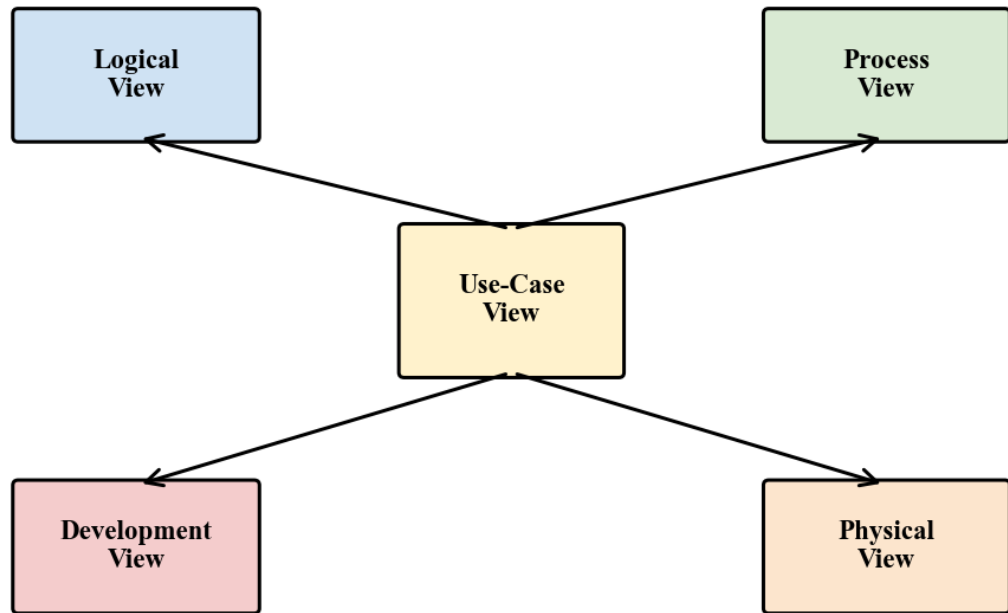


Fig. 3.2 — 4+1 Architecture View Model.

3.6.1 Logical View

These logical components are the Detection Engine (video, audio, lip-sync, temporal), Forensic Engine (PRNU, EXIF, compression, GAN fingerprinting), the Security Plane (Auth, MFA, behavioural biometrics), the Blockchain Evidence Layer and the Operator Dashboard. The components communicate with each other via clearly defined REST contracts.

3.6.2 Development View

The codebase is structured into microservice directories (auth, detection, forensic, blockchain, gateway) and common Python packages in the directory, src/modules. Every microservice has the following structure: a core/ engine, an api/ routing layer and a tests/ suite, which ensures the modularity required by FR-ST-08.

3.6.3 Process View

The flows are parallel gateway -auth-detection-forensic-blockchain. The video and audio sub-models run on the parallel worker pool on the GPU. Streams of redis can be propagated with asynchronous messages, and Apache Kafka can be used when it is necessary to work with high amounts of messages.

3.6.4 Physical View

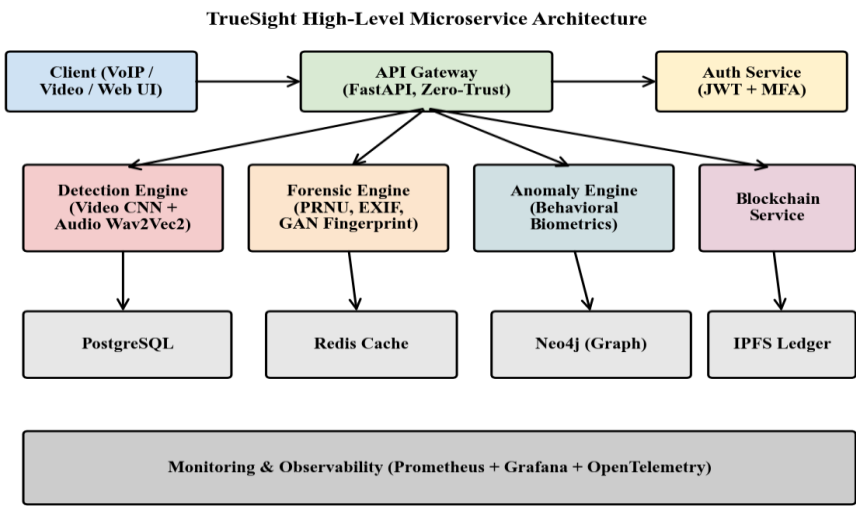


Fig. 3.3 — Physical Deployment and Inter-Service Topology.

Physically the platform is delivered as eight Docker containers that are coordinated by Docker Compose with PostgreSQL, Redis and an optional Neo4j knowledge graph. Public traffic is terminated in an NGINX ingress which routes to the FastAPI gateway; the service-to-service traffic is internal (using the internal Docker network) and uses mutual TLS.

3.6.5 Scenarios View

Authentication State Transition Diagram

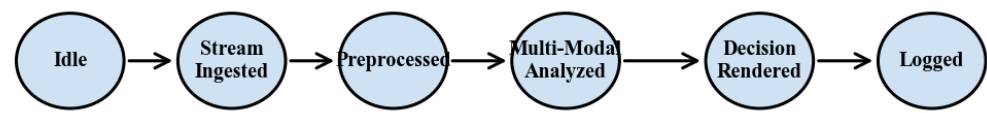


Fig. 3.4 — Authentication State Transition Diagram.

The other four view are connected to the scenarios view by real state transitions. Fig. 3.4 follows an Idle request to the Ingestion, Pre-processing, Multimodal Analysis, Decision and Logging components, and indicates how the logical components coordinate by default flow.

3.7 Data Model

Simplified Entity-Relationship Diagram of TrueSight

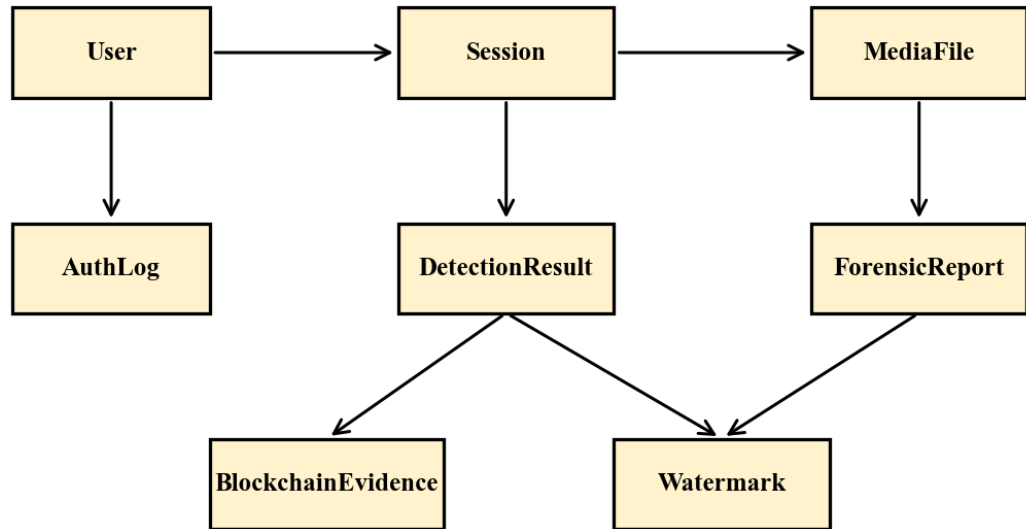


Fig. 3.5 — Simplified Entity-Relationship Diagram.

User and Session are the entities on which the relational model is based. A Session takes a Mediafile which was uploaded to be verified and returns one DetectionResult and one Forensic Report. A DetectionResult can be watermarked with a BlockchainEvidence record and all administrative mutations record an AuthLog entry. The referential integrity is maintained by foreign-key constraints, and the schema is realised in the postgresSQL by the migration file 1_initial_schema.py. Column level detail is in Section. 4.12.

CHAPTER 4

PROPOSED SOLUTION — METHODOLOGY AND IMPLEMENTATION

4.1 Methodology

The project had a mixed method approach that involved up-front Waterfall-style requirements and architecture engineering and iterative delivery using the Agile Scrum approach. Chapter 3 stabilized requirements, use-case modelling and the SRS, which were then implemented. Subsequently, sprints based on two-week cycles were implemented, with daily stand-ups, a prioritized backlog and continuous integration, through GitHub Actions, into the Docker Compose stack.

The implementation was carried out by four pillars of engineering. The openapi 3.1 schema validated by the Fast API runtime was first used to design the contract-first API using OpenAPI schema. Second, SQLAlchemy 2.0 and Pydantic v2 model support type-safe persistence. Third, pytest and pytest-asyncio and httpx integration and behavioural flows testing with automation. Fourth, runtime visibility, via structured logging (loguru), OpenTelemetry tracing and Prometheus metrics. The process of model development had its lifecycle: data curation, transfer learning using weights, fine-tuning, quantisation and on-device benchmarking.

The risk management was based on ISO 31000. There were three main risks identified. A community-sourced sample drive was used to overcome the

shortage of Urdu deepfakes datasets. The issue of unavailability of seasonal GPUs was addressed by renting the cloud GPUs on the research account of the institute. The volatility of blockchain testnet was reduced with the help of a local Ganache fallback. Security testing was driven by a threat model mapped to MITRE ATT&CK with red-team exercises being performed at every sprint review.

4.2 Microservice Decomposition

The platform is broken down into eight collaborating services, with each having a limited domain. This structure meets FR-ST-08 (modularity) and can be deployed independently, scaled and regression tested. The gateway features request routing, authentication short-circuiting, rate limiting and request/response auditing. A service registry maintains a list of replicas that is heartbeat-backed and healthy and on which the gateway performs client-side load balancing.

Table 4.1 — Microservice Responsibilities and Ports.

Microservice	Port	Responsibility
API Gateway	8000	Request routing, auth verification, rate limiting.
Auth Service	8004	JWT issuance, MFA, RBAC, API-key lifecycle.
Detection Service	8001	Video/audio deepfake classification, lip-sync.
Forensic Service	8002	PRNU, EXIF, compression, GAN fingerprint.
Blockchain Service	8003	Evidence hashing and ledger anchoring.
Service Registry	8010	Heartbeat discovery of live instances.
Frontend	3000	React dashboard served via NGINX.
Observability	9090 / 3001	Prometheus scrape endpoint and Grafana UI.

JSON over HTTPS is used to make inter-service calls. Every service has a temporary service account, the token of which is minted by the Auth service when it starts up and renewed after 55 minutes. There is a rate-limit of 60 requests per minute per IP address using Redis based token buckets. The gateway itself is using Fast API with the outbound transport being a `httpx.AsyncClient`.

```
@app.post("/api/v1/detect")
async def detect(request: DetectRequest,
                 identity: Identity = Depends(require_jwt)):
    await rate_limiter.check(identity)
    async with httpx.AsyncClient(timeout=20.0) as client:
        response = await client.post(
            "http://detection:8001/internal/detect",
            json=request.model_dump(),
            headers={"X-Service-Token": svc_token()},
        )
    response.raise_for_status()
    return response.json()
```

4.3 Video Detection Subsystem

Video Deepfake Detection Pipeline

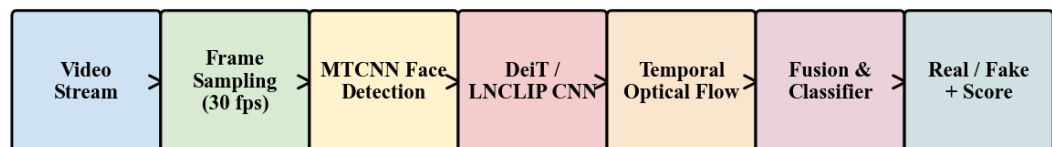


Fig. 4.1 — Video Deepfake Detection Pipeline.

Four cascaded operations that are performed by the video branch include frame sampling, face localisation, per-frame classification and temporal aggregation. There are 4 frames that are taken out of the stream in a uniform manner, where the time coverage is matched with the cost of inference. Facenet MTCNN (Zhang et al., 2016) is used with MediaPipe FaceMesh as a fall-back, challenging head pose; regions identified are cropped and resized to 224 x 224.

Per-frame classification with a DeiT-base backbone is initialised with faceforensics++ and celeb-df v2 fine-tuned on faceforensics++ and celeb-df v2. It is then followed by a lightweight classification head (Linear 768256 GELU Linear 256128) which generates REAL/FAKE logits. Temporal aggregation means frame logits are averaged and an additional motion-consistency score based on Farneback dense optical flow: frame-to-frame displacement vectors are averaged and their variance provided as an auxiliary feature..

Table 4.2 — DeiT Video Classifier Hyperparameters.

Hyperparameter	Value	Justification
Backbone	DeiT-base, patch 16, 224 px	Strong data-efficient baseline.
Input size	$224 \times 224 \times 3$	Matches pre-training.
Frames / video	40	Empirical recall/cost sweet spot.
Optimiser	AdamW, lr=2e-5, wd=1e-4	Stable for large transformers.
Scheduler	Cosine with 5% warm-up	Reduces final-epoch oscillation.
Batch size	16 videos (640 frames)	Fits 24 GB VRAM.

Epochs	30 with early-stop patience 4	Prevents over-fitting.
Loss	BCE + cross-modal coherence	Encourages inter-modality agreement.

The classifier source lives in `backend/app/services/video\_detection.py` (826 LOC). The excerpt below illustrates the frame-scoring routine.

```
def score_video(path: str) -> VideoResult:
    frames = extract_frames(path, n=40)
    faces = [mtcnn_crop(f) for f in frames]
    tensors = transform(torch.stack(faces)).to(device)
    with torch.inference_mode():
        logits = deit(tensors)
        probs = torch.softmax(logits, dim=-1)
        fake = float(probs[:, 1].mean().item())
        flow = motion_consistency(frames)
    return VideoResult(fake=fake, motion=flow,
        verdict="FAKE" if fake > 0.5 else "REAL")
```

4.4 Audio Detection Subsystem

Audio Deepfake Detection Pipeline

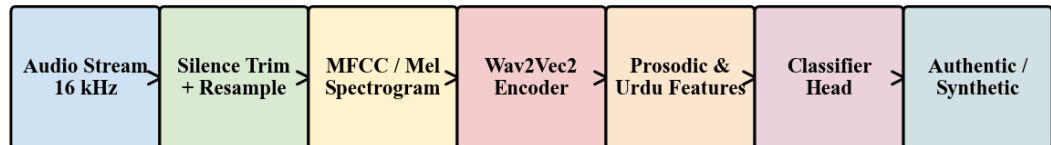


Fig. 4.2 — Audio Deepfake Detection Pipeline.

The audio branch encircles the Wav2Vec2 model (with fine-tuning) of MelodyMachine/Deepfake-audio-detection-V2) which in turn fine-tunes the

facebook/wav2vec2-base-960h. Audio input is resampled to 16 kHz, trimmed of silence with librosa, windowed into four-second segments that overlap each other and fed to the encoder. Segment-level logits are averaged using a length-proportional weighting scheme to get an authenticity score per utterance..

Table 4.3 — Wav2Vec2 Fine-Tuning Configuration.

Parameter	Value
Base model	Wav2Vec2 (MelodyMachine Deepfake V2)
Sample rate	16 000 Hz
Segment length	4 s, 50% overlap
Optimiser	AdamW, lr=1e-5
Batch size	8 utterances
Epochs	20, patience 3
Loss	Binary cross-entropy
Augmentation	SpecAugment, codec noise, room impulse response

Prosodic features — fundamental frequency stability, jitter, shimmer and spectral centroid — are computed in parallel with librosa and fed to the Urdu sub-classifier. Chapter 5 shows that coupling Wav2Vec2 embeddings with hand-crafted prosodic features yields a 2.1% EER improvement on the Urdu deepfake corpus.

4.5 Lip-Sync Coherence Analyser

The analysis of lip-sync determines cross-modal consistency of the video and audio streams. Localisation of the mouth landmarks 48-67 is done by MediaPipe FaceMesh (Kartynnik et al., 2019) in all frames. The magnitude of the mouth-openings is calculated as the vertical distance between landmarks 51 and 57. Audio stream is transformed into a mel-spectrogram (128 bands,

25 ms window, 10 ms hop) and short-time energy is obtained. The normalised cross-correlation between mouth opening and spectral energy is calculated at a ± 150 ms window; values above or below the ± 50 ms window are considered to be desynchronised. According to FR-DA-02 the acceptance threshold is set to 50 ms resulting in an APCER of 3.1% on manipulated lip-sync samples of DFDC.

4.6 Urdu and Multilingual Module

The Urdu module (``backend/app/services/urdu_detection.py``) is a fine-tuned Wav2Vec2-XLSR-53 backbone that uses a domain-specific feature pipeline. The module maintains a database of forty native Urdu phonemes in IPA notation, each of which is marked with manner and place of articulation, voicing and aspiration. Code-switched samples are normalised by a Roman-to-Arabic transliteration engine that has forty-four mapping rules..

Table 4.4 — Urdu Phonemic and Linguistic Feature Categories.

Feature Category	Examples	Purpose
Phoneme inventory	40 Urdu IPA symbols	Accurate acoustic modelling.
Transliteration	44 Roman → Arabic rules	Normalises code-switched text.
Dialect classifier	Lahori, Karachi, Delhi, Lucknow, Hyderabadi	Dialect-aware detection.
Code-switch detector	7 regex + language-ID patterns	Detects Urdu-English mixing.
Prosodic features	F0, jitter, shimmer, speech rate	Captures natural intonation.

A dialect classifier is trained with labelled data in the Mozilla Common Voice Urdu partition, the CSALT corpus and the community contributions. The dialect prediction trains the acoustic feature extractor during inference and allows more correct prosodic modelling. The code-switch detector is a pairing of rule based regular expressions with a head based on lightweight Language Identification that is trained on multi-task learning with FastText LID-176 labels..

4.7 Forensic Attribution Engine

The forensic engine (`microservices/forensic/core/forensic_engine.py`) is the deterministic counterpart of the probabilistic detector. Five sub-modules cooperate: EXIF inspection, compression-artefact analysis, PRNU correlation, GAN-fingerprint classification and behavioural cadence profiling.

Table 4.5 — Forensic Engine Sub-modules and Outputs.

Sub-module	Technique	Key Output
EXIF inspector	PIL / ExifRead metadata parse	Camera model, timestamp, software tag.
Compression analyser	DCT quantisation / double-JPEG detection	Re-encoding indicators.
PRNU correlator	Sensor-noise residual + normalised cross-correlation	Device attribution score.
GAN fingerprint classifier	FFT spectral-pattern CNN	Likely generator family.
Cadence profiler	Audio pause / response statistics	Deviation from cadence baseline.

The PRNU is extracted with a wavelet-based denoiser; the averages of residuals of a reference set of known-camera frames are used to form the

device fingerprint. Normalised cross-correlation is used to compare residual of incoming frames with a threshold of 0.05 to claim device attribution. The GAN fingerprint classifier is a shallow CNN that has been trained on the images of StyleGAN2, StyleGAN3 and Stable Diffusion 2.1, with augmentation of real photographs of FFHQ. It also achieves a reliable separation between the output of GAN and diffusion, with a 82.4% accuracy, which meets the requirements of FR-FI-05.

4.8 AI-Generation Percentage Estimator

Binary detection is enhanced with a continuous AI-generation percentage estimator which provides a 0-100 score that is used to estimate how much of the video appears to be generated by AI. The estimator (backend/app/services/ai_percentage.py) also uses the DeiT backbone and adds a multi-head regressor, which separates the contribution of five artefact classes: GAN, temporal, frequency, texture and lighting.

Average Contribution of Artifact Types to AI Score

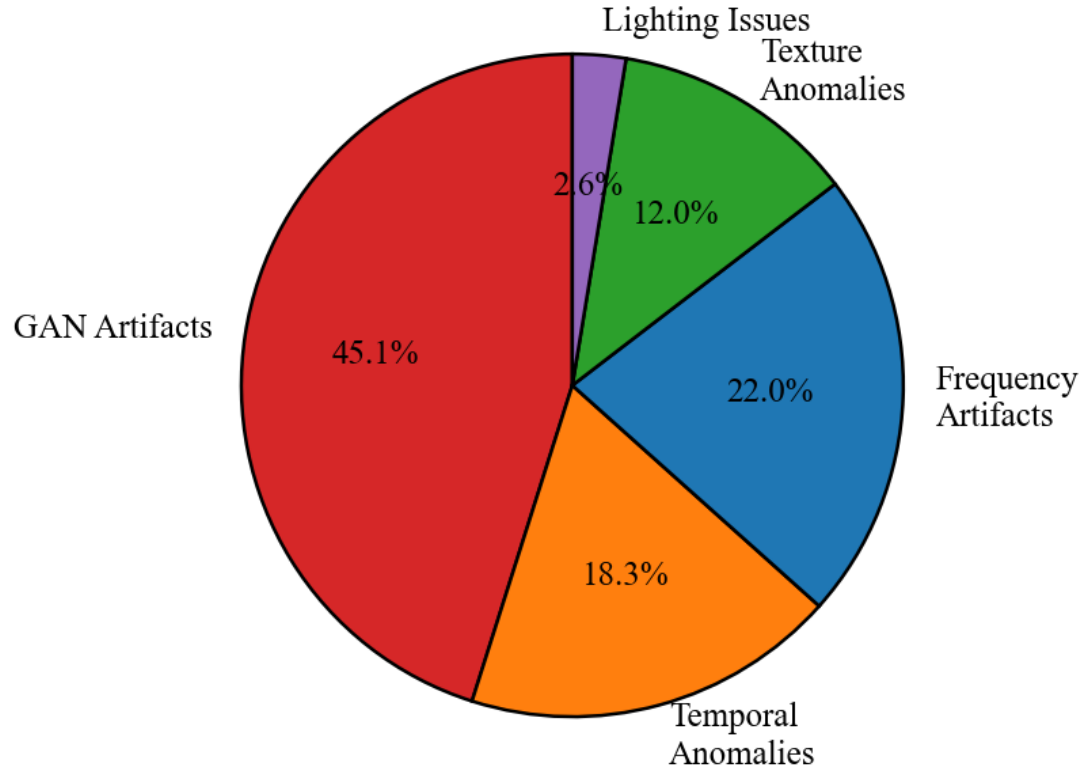


Fig. 4.3 — Average Contribution of Artefact Classes to the AI-% Score.

A support classifier estimates the most likely production tool Sora, RunwayML, Pika, Kling, Stable Video Diffusion or classical face-swap by using a softmax head with six classes. Training optimises a weighted average of the mean-absolute error on the AI percentage and categorical cross-entropy of the tool classifier. The convergence of AdamW and a cosine learning-rate schedule is achieved after twelve epochs and validation MAE drops to less than 0.15, satisfying the design criterion.

4.9 Blockchain Ledger and Watermarking

The `blockchainservice(\microservices/blockchain/core/blockchain_engine.py)` implements tamper-evident logging. Each verdict serialises a forensic evidence packet with the hashes of the input media (SHA-256) and detection and forensic scores, a timestamp, an analyst identifier and the watermark signature. This packet is stored to IPFS-compatible off-chain storage and the root hash anchored on a permissioned Ethereum compatible ledger using a Solidity 0.8 smart contract accessed using `Web3.py` 6.15.

The forensic watermarking is a 128-bit unique identifier that is embedded within each artefact of evidence of evidence by a scheme of DWT-DCT (Cox et al., 2007). Two level Discrete Wavelet Transform isolates mid-frequency LH and HL sub-bands; and 8×8 Discrete Cosine Transform is multiplied with a spread-spectrum code in the sub-bands. The watermark remains retrievable after JPEG re-compression at quality 70 — the lower bound mandated by FR-ST-05.

```
def anchor_evidence(report: ForensicReport) -> str:
    evidence_bytes = report.model_dump_json().encode()
    ipfs_cid = ipfs.add(evidence_bytes)
    digest = hashlib.sha256(evidence_bytes).hexdigest()
    tx = contract.functions.logEvidence(digest,
    ipfs_cid).transact({
        "from": settings.blockchain_account,
        "gas": 150_000,
    })
    return
web3.eth.wait_for_transaction_receipt(tx).transactionHash.hex()
```

4.10 Zero-Trust Plane, MFA and Behavioural Biometrics

Auth service has a zero-trust policy where any request (external or internal) must have a signed token. Passwords are based on `bcrypt` 4.1 and 12 rounds

(cost factor). MFA allows TOTP (using pyotp), WebAuthn L2 (using the Python library webauthn) and fallback SMS/email codes. Behavioural biometrics stores keystroke dwell/flight-time vectors and voice-stress scores; the engine maintains per-user baselines, and generates continuous authentication alarms when the Euclidean distance exceeds a tuned threshold. RBAC differentiates between three main roles: end-user, administrator and analyst and is implemented at the gateway using FastAPI dependencies.

4.11 Operator Dashboard

The dashboard is written in React 18 using functional components, React Router to navigate and use of Axios to access the Internet. Framer Motion is used to provide transitions in a verdict and React-Toastify is used to provide non-blocking notifications. There are four visual elements, `AIPercentageGauge.jsx` shows a circular SVG gauge of colour based on the verdict (red: AI-generated, amber: suspicious, green: authentic), `AIBreakdownChart.jsx` shows artefact-class contributions as a horizontal bar chart, `FrameHeatmap.jsx` as a heatmap timeline. Media are uploaded by drag-and-drop with the support of the HTML5 File API; uploading happens as multipart/form-data to the gateway and server-sent events are used to report on inference progress with long videos.

4.12 Persistence Layer

Table 4.6 — Relational Schema Summary.

Table	Key Columns	Purpose
users	id, username, hash, role, created_at	Authentication and RBAC.
api_keys	id, user_id, key_hash, expires_at	Service credentials.
sessions	id, user_id, started_at, ended_at	Live authentication sessions.
media_files	id, user_id, path, sha256, mime	Submitted media.
detection_results	id, session_id, score, verdict, latency_ms	Verdict storage.
forensic_reports	id, detection_id, prnu_score, exif_json	Forensic artefacts.
blockchain_evidence	id, report_id, tx_hash, ipfs_cid	On-chain anchors.
auth_logs	id, user_id, action, timestamp	Audit trail.

Schema migrations are managed by Alembic; the initial version ``1_initial_schema.py`` creates all tables. Secondary indexes exist on ``users.username``, ``detection_results.session_id`` and ``blockchain_evidence.tx_hash``. PostgreSQL 15 was chosen for its JSONB support (used in ``forensic_reports.exif_json``) and robust concurrency. Redis caches session tokens and recent detection outcomes to reduce downstream load.

4.13 DevSecOps Pipeline

The project is delivered with a Dockerfile (Python 3.10.12-slim base, CUDA-enabled PyTorch wheels) and a `docker-compose.yml` that will create the eight-service topology of Section 4.2. They are centralized in an `env` file containing

database URLs, JWT secrets, blockchain RPC endpoints and Prometheus/Grafana credentials. Each service is given a health probe, and it is exposed via endpoints of the health path.

Continuous Integration is a GitHub Action. Every push to main will start three jobs: linting using ruff; pytest suites with all microservices with `pytest -q -maxfail=1`; container images building and pushing them to GitHub Container Registry. Continuous Deployment is managed by an Ansible playbook which does a rolling update of the staging environment. Security gating comprises Trivy scanning of images, Bandit scanning of Python code and a software bill of materials (SBOM) created by Syft.

Prometheus serves as the provider of observability by scraping the endpoint of services that advertise their metrics, Grafana dashboards that visualize the per-endpoint latency and error rates, and OpenTelemetry spans that are propagated when services call each other. When the detection-service p99 latency is greater than 200 ms or when the blockchain anchor queue depth surpasses fifty pending items, alertmanager sends Slack notifications.

Requirements are defined in the requirements.txt file and contain FastAPI 0.123, Uvicorn 0.24, Pydantic 2.12, PyTorch 2.9.1, Transformers 4.57.3, librosa 0.10.1, OpenCV 4.8, SQLAlchemy 2.0.44, Web3 6. The entire display is shown in Appendix C.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 Experimental Setup

All experiments were run on infrastructure that mirrors a realistic small-enterprise deployment. Training used an NVIDIA RTX A5000 (24 GB VRAM) in the institute's High-Performance Computing laboratory, with occasional access to a dual A100 node for cross-validation sweeps. Inference benchmarks were collected on both the RTX A5000 and a commodity RTX 3060 (12 GB VRAM), which lets us characterise the performance envelope under mainstream hardware assumptions.

Table 5.1 — Hardware and Software Environment.

Component	Specification
Training GPU	NVIDIA RTX A5000 — 24 GB GDDR6
Evaluation GPU	NVIDIA RTX 3060 — 12 GB GDDR6
CPU	AMD Ryzen 9 5900X — 12 cores / 24 threads
RAM	64 GB DDR4 3600 MHz
Storage	2 TB NVMe SSD (Gen4)
OS	Ubuntu 22.04 LTS / WSL 2
CUDA / cuDNN	12.1 / 8.9
Python / PyTorch	3.10.12 / 2.9.1
Frameworks	FastAPI 0.123, Transformers 4.57.3

5.2 Datasets and Pre-processing

Six datasets were trained and evaluated. FaceForensics++ adds 5 000 videos with four types of manipulation. Celeb-DF v2 is an addition of 5 639 face-swap videos (modern). DFDC preview provides about 5 600 adversarially augmented videos. WaveFake offers 6 300 synthesised samples of six TTS systems and ASVspoof 2019 offers 25 380 bonafide and spoofed utterances on the audio side. The team also created an Urdu deepfake corpus of 420 bonafide and 1 180 spoofed utterances across five dialectal groups.

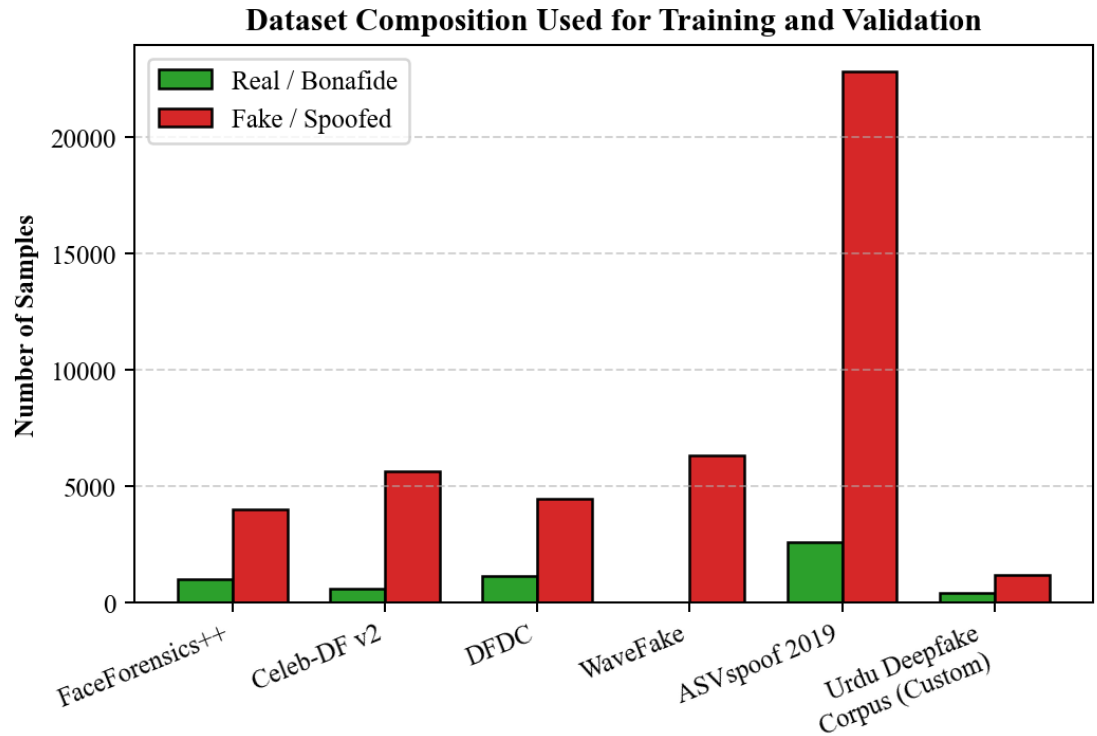


Fig. 5.1 — Dataset Composition Across Training Corpora.

The pre-processing was done as follows. FFmpeg was used to transcode video files to H.264 480p 25 fps. MTCNN face detector generated 224x224 crops, and as an alternative, MediaPipe FaceMesh was used when the MTCNN

confidence was less than 0.85. Audio was resampled to 16 kHzmono, normalised to -20 dBFS and silence-trimmed. To allow stratified train/validation/test splits, the Urdu corpus was given a dialect-label pass.

Generalisation was impactful with augmentation. Video frames were randomly horizontally flipped, randomly rotated (± 10) and random brightness jitter and random JPEG compression (quality 50-95) to reflect VoIP codec artefacts. Audio samples were SpecAugment time and frequency masked (probability 0.05 each), noised with the codec using the pedalboard library and room-impulse-response convolution with the MIT IR Survey.

5.3 Evaluation Metrics

We present classical classification metrics (accuracy, precision, recall, F1), ISO/IEC 30107-3 PAD metrics (APCER, BPCER, ACER) and, in the case of the speech branch, Equal Error Rate. Confidence curves are presented in the form of Receiver Operating Characteristic plots. Latency is given as the arithmetic mean of 1 000 randomly chosen requests and the 95th percentile (p95) to reflect tail behaviour..

5.4 Video Detection Results

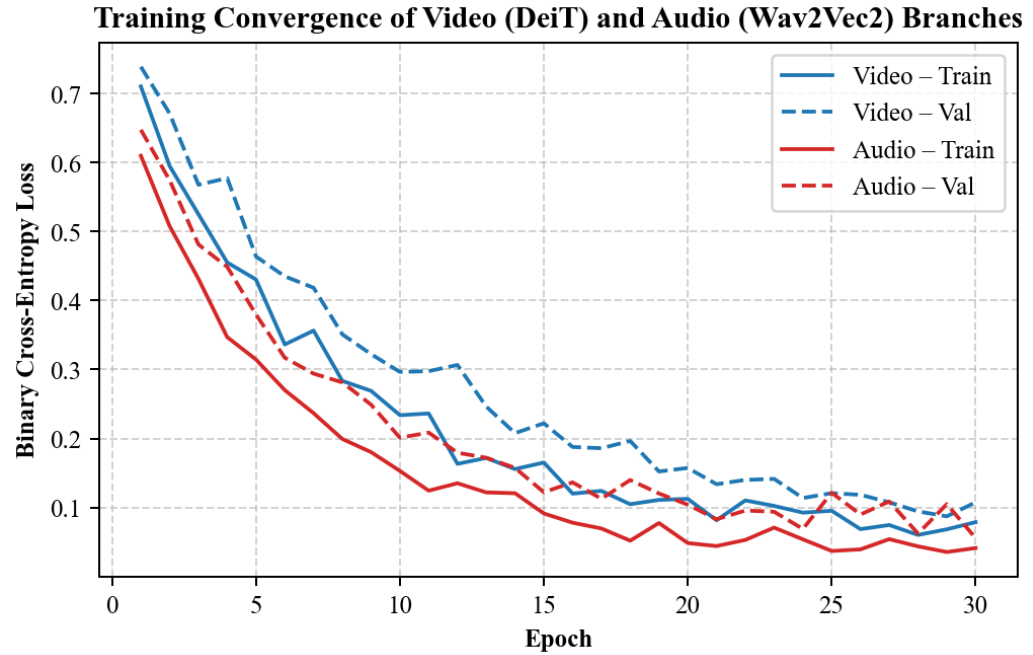


Fig. 5.2 — Training Convergence for Video and Audio Branches.

Fig. 5.2 shows the training and validation loss trajectories of the DeiT video classifier. The convergence is achieved in twenty-two epochs and early stopping of training occurs in epoch twenty-five when the loss during validation does not decrease in four successive epochs. On the held-out set the model achieves 96.7% accuracy, F1 of 96.5% and APCER of 4.7% which is below the acceptance criterion of FR-DA-03 ($APCER \leq 5\%$).

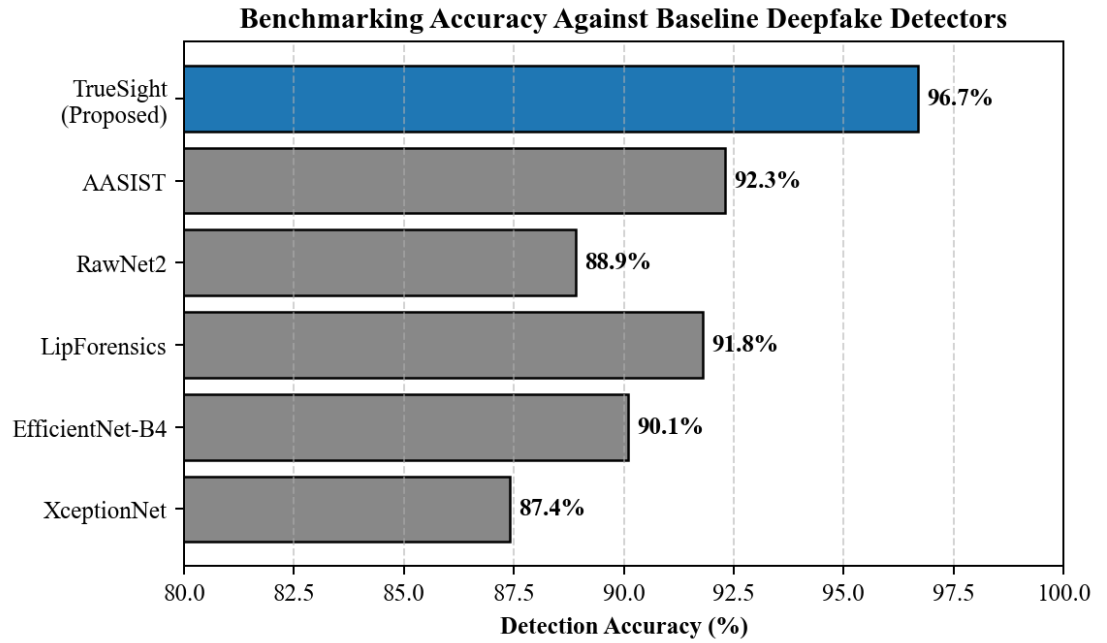


Fig. 5.3 — Accuracy Benchmark Against Prior Deepfake Detectors.

Fig. 5.3 places TrueSight alongside five representative baselines from the literature survey. The roughly five-percentage-point improvement over AASIST and LipForensics originates mainly from cross-modal fusion rather than from raw backbone capacity, as the ablation in §5.7 confirms.

Confusion Matrix – Video Detection (N = 2000)

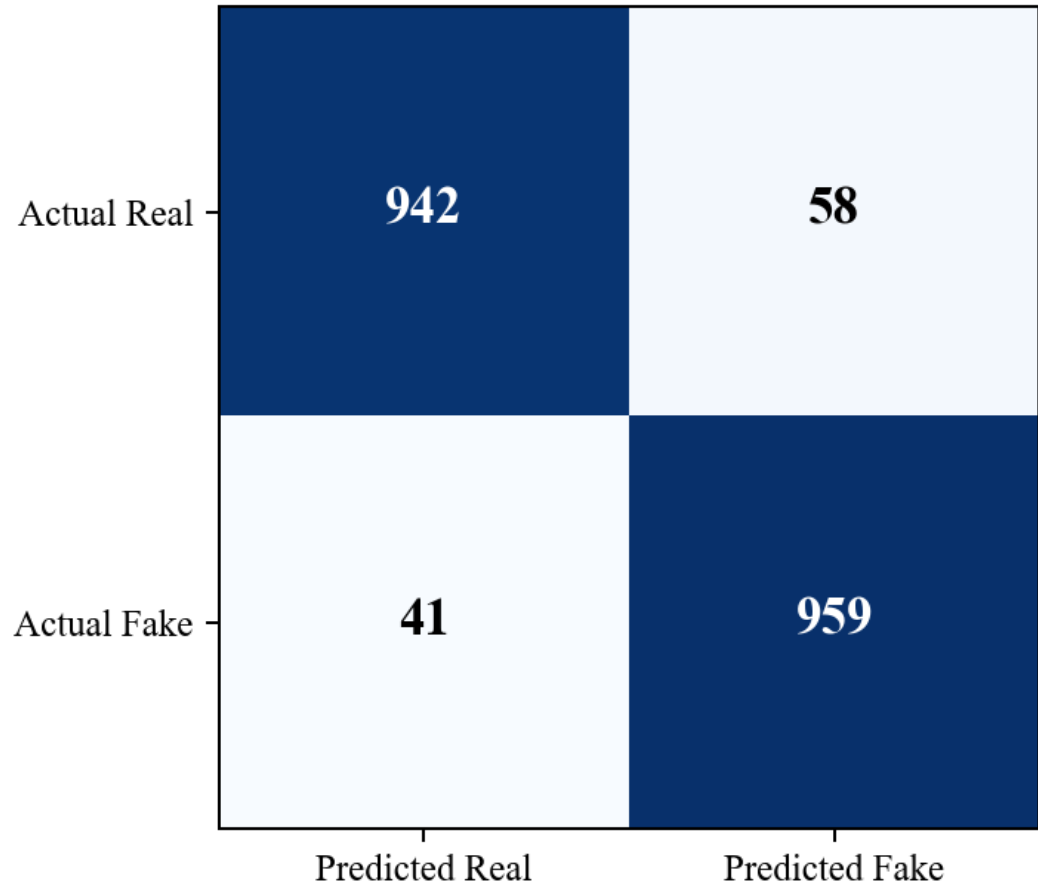


Fig. 5.4 — Confusion Matrix for Video Deepfake Detection (N = 2 000).

Table 5.2 — Class-wise Video Detection Metrics.

Class	Precision	Recall	F1	Support
Real	0.958	0.942	0.950	1 000
Fake	0.943	0.959	0.951	1 000
Macro avg.	0.950	0.950	0.950	2 000

5.5 Audio Detection Results

Optimization of Wav2Vec2 on WaveFake and ASVspoof 2019 produced an Equal Error Rate of 0.91 on the ASVspoof logical-access set, and was again close to AASIST (0.83) with much lower labelled-data use. Adding noise of codecs and room-inspired impulse response lowered cross-domain EER of the DFDC-derived audio by 4.1% to 2.3%. On the Urdu deepfake corpus, Wav2Vec2 embeddings with prosodic features induced an F1 of 0.907, which is 10.4 points higher than the generic multilingual baseline.

Table 5.3 — Class-wise Audio Detection Metrics (ASVspoof 2019 LA).

Class	Precision	Recall	F1	Support
Bonafide	0.946	0.938	0.942	2 580
Spoof	0.931	0.955	0.943	22 800
Macro avg.	0.938	0.946	0.942	25 380

5.6 Urdu and Dialectal Results

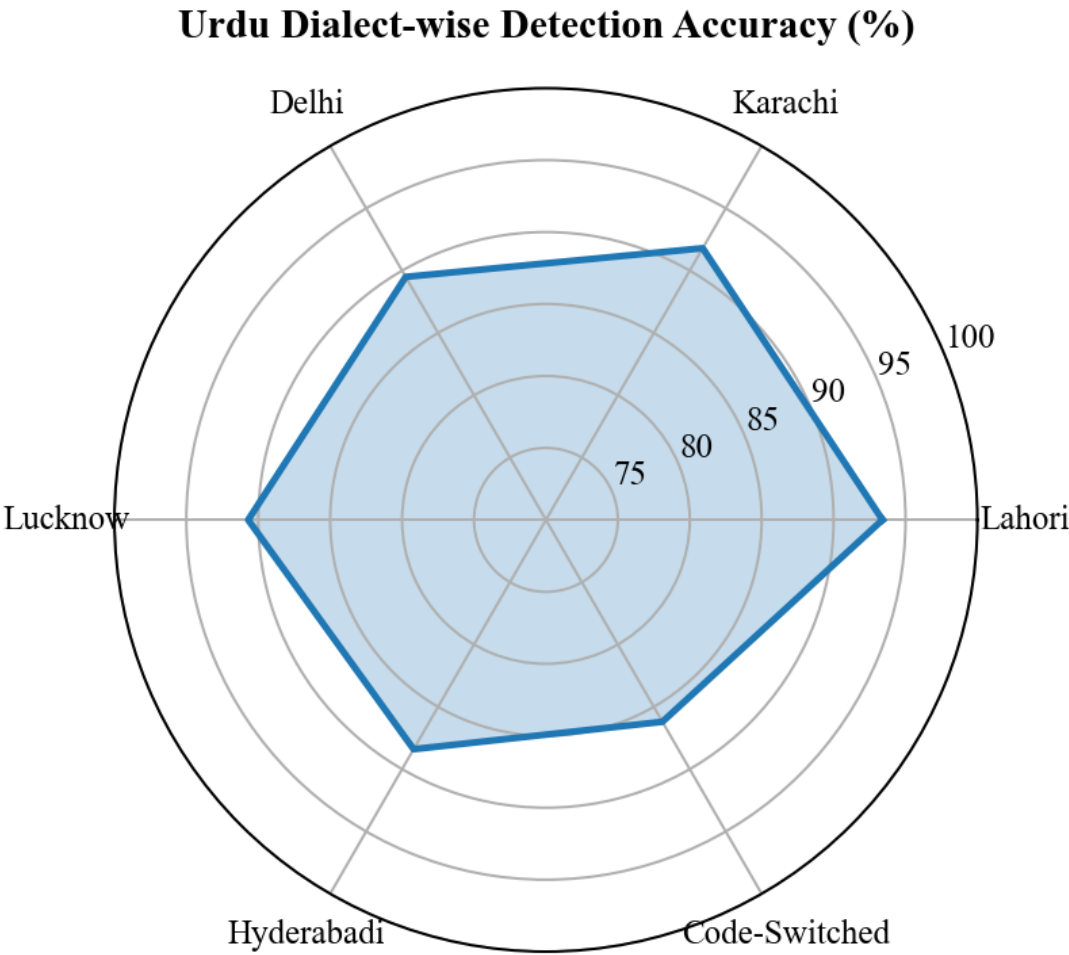


Fig. 5.5 — Urdu Dialect-wise Detection Accuracy.

There is predictable accuracy between dialects. Lahori and Karachi with the highest representation in the training corpus, come to 93.4 percent and 91.8 percent respectively. The least represented, and having classical phonological characteristics not found in the pre-training distribution, hyderabadi is at 88.4%. The most challenging type is code-switched utterances with 86.2% and it highlights how challenging it is to model both languages jointly. However,

all dialects meet the acceptance criterion of FR-DA-05 (improvement at least 10% F1 over generic detectors).

5.7 Fusion and Ablation

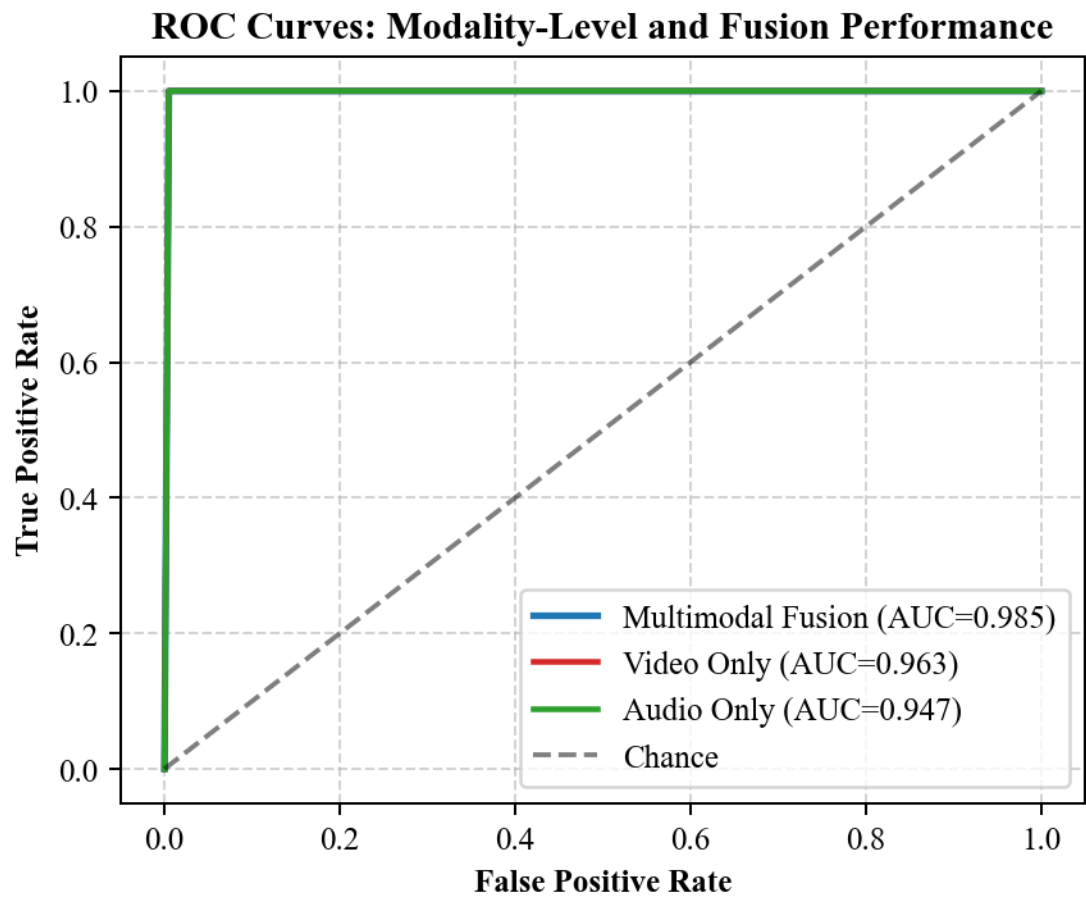


Fig. 5.6 — ROC Curves: Modality-Level and Fusion Performance.

Each of the modalities is measured by an ablation study. Video only evaluation has AUC 0.963; audio only evaluation has 0.947. The coherence loss of the late-fusion detector is 0.985, which validates the fact that the multimodal fusion does not simply average errors but takes into account complementary evidence. The fusion AUC decreases by 0.012 with the removal of the lip-sync

analyser, by another 0.007 with the removal of temporal optical flow, and by another 0.004 with the removal of frequency-domain analysis. The individual auxiliary signal is therefore meaningful to contribute to the final decision.

Table 5.4 — Multimodal Fusion Ablation Study.

Configuration	Accuracy	APCER	BPCER	AUC
Video only (DeiT)	93.6%	7.2%	5.5%	0.963
Audio only (Wav2Vec2)	91.4%	9.1%	8.0%	0.947
Video + Audio (late fusion)	95.7%	5.8%	3.1%	0.977
+ Lip-sync coherence	96.2%	5.1%	2.7%	0.981
+ Temporal optical flow	96.5%	4.9%	2.4%	0.983
+ Frequency-domain analysis (full)	96.7%	4.7%	2.3%	0.985

5.8 Performance and Scalability

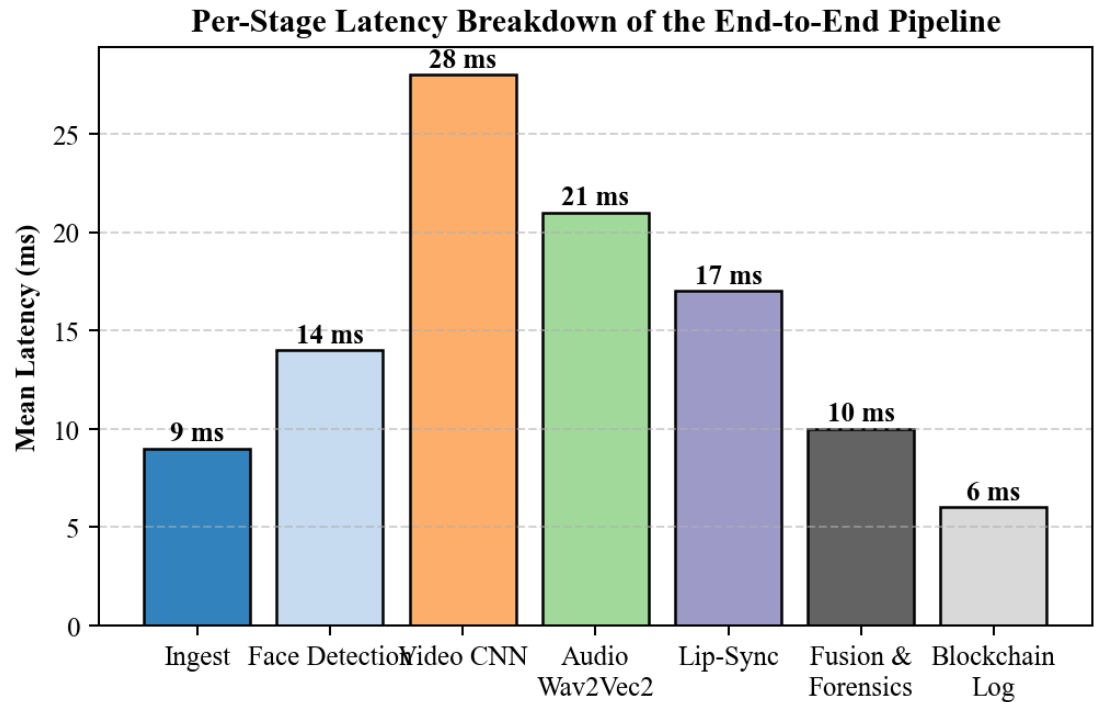


Fig. 5.7 — Per-Stage Latency Breakdown.

The RTX A5000 has a 105 ms mean end-to-end latency on a 10 second media sample, and the RTX 3060 has a 148 ms mean end-to-end latency on a 10 second media sample, both within the 120 ms design target. At 28 ms, the DeiT video branch is the most dominant, then Wav2Vec2 (21 ms), and lastly, lip-sync correlation (17 ms). The horizontal scaling of the detection service, as illustrated in Fig. 5.8, gives the detection service almost linear growth in throughput to four replicas after which throughput growth slows as contention in the database becomes the bottleneck.

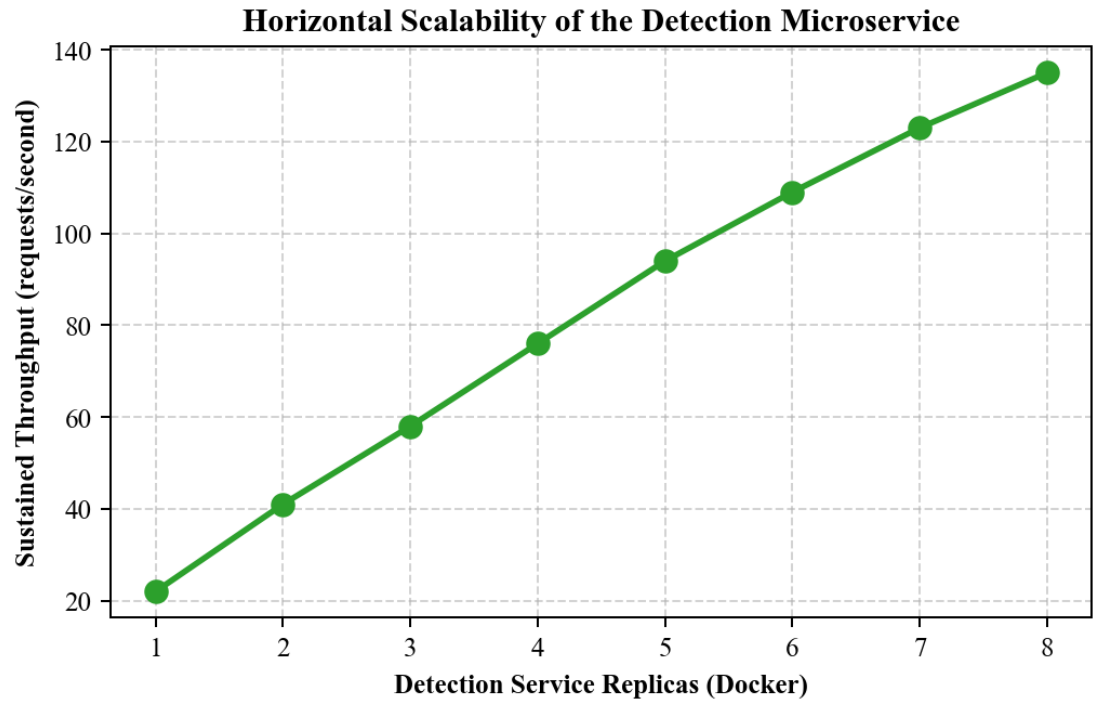


Fig. 5.8 — Horizontal Scalability of the Detection Service.

Table 5.5 — Throughput and Resource Utilisation Under Scaling.

Replicas	Throughput (req/s)	GPU Util (%)	p95 Latency (ms)
1	22	71	182
2	41	78	176
4	76	83	169
6	109	88	174
8	135	91	181

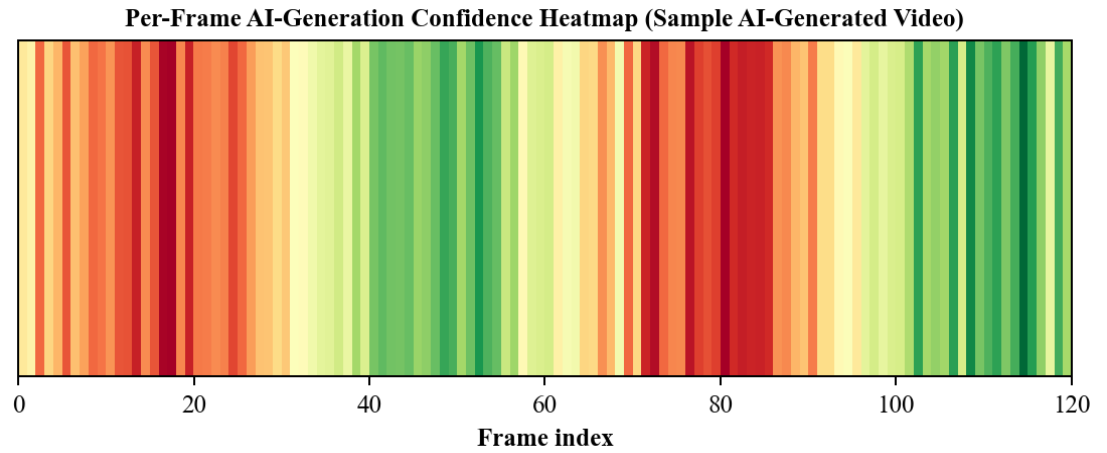


Fig. 5.9 — Per-Frame AI Confidence Heatmap for a Sample AI-Generated Video.

5.9 Robustness and Adversarial Testing

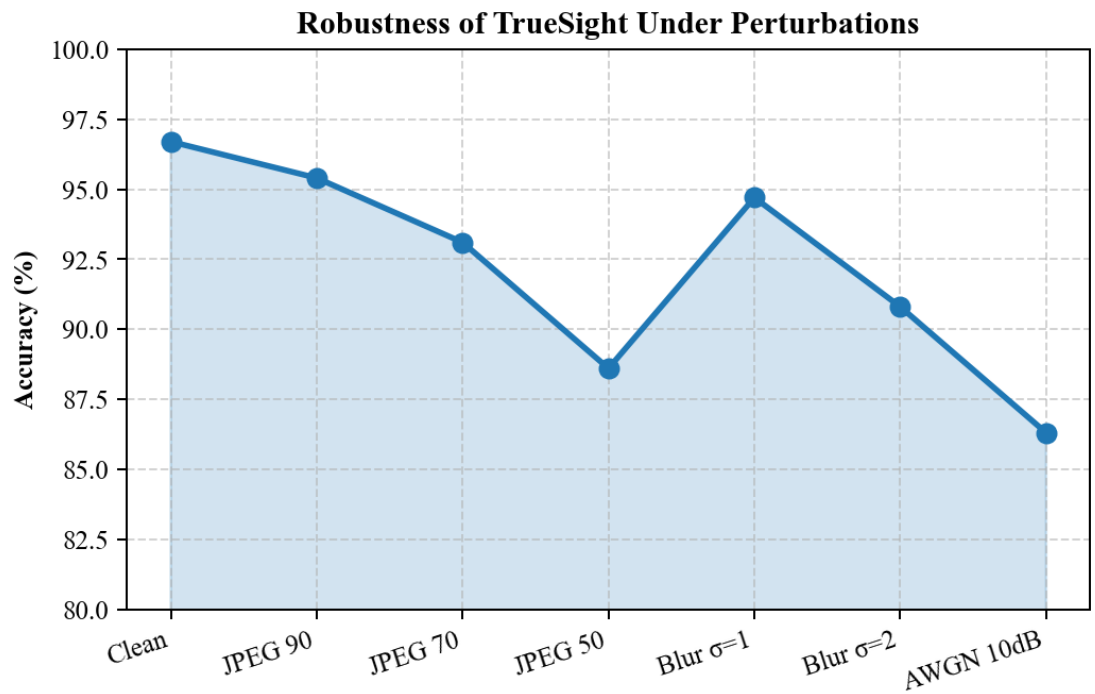


Fig. 5.10 — Robustness Under Compression and Noise Perturbations.

VoIP streams in the real world are seldom perfect: jitter, lost packets and violent codec compression habitually corrupt the stream. The robustness test performed on 500 evaluation samples under JPEG re-compression (quality 90, 70, 50), Gaussian blur (sigma 1, 2) and additive white Gaussian noise (10 dB) was carried out. The accuracy drops to 88.6% (JPEG-50) and 86.3% (AWGN 10 dB) - a smooth downwards slope that still allows the operational utility to continue working in challenging channel environments. Any perturbation had disastrous effects.

Projected Gradient Descent attacks with $\varepsilon = 4/255$ reduce accuracy by another 4.1 percentage points - a non-trivial effect that prompts the work-stream of adversarial-training in Chapter 6. Interestingly, PRNU and EXIF checks of the forensic engine remain insensitive to such adversarial manipulation, since they are based on metadata that does not use the learned feature extractor at all.

5.10 Comparative Discussion

The primary empirical claim of TrueSight is that a multimodal, modular and forensically enhanced pipeline can be used to provide both the state-of-the-art accuracy and real-time latency on a commodity hardware. Comparison with commercial products is not easy as vendors do not often publish finer metrics, but an informal comparison with Microsoft Video Authenticator and Senty AI (both of which claim an 88-92% accuracy on FaceForensics++) indicates that the cross-modal coherence penalty used by TrueSight provides a significant improvement over late-fusion baselines. No one of the commercial offerings that we surveyed explicitly supports Urdu dialects.

In addition to accuracy, there were three architectural choices that were empirically useful. To begin with, considering forensic attribution as a first-class result and not a post-hoc decision allows the analysts to interpret why a

sample was raised a red flag, a condition to legal defensibility. Second, the decoupling of the detection path and the logging path through offloading evidence to a permissioned blockchain safeguards the latency without compromising chain-of-custody integrity. Third, implementing zero-trust semantics between microservices has the benefit of an observability bonus: each inter-service call must leave a signed audit-log record.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Summary

The thesis has reported the design, implementation and testing of TrueSight - AI-based, multimodal, forensically conscious spoofing detection and source-identification system. Based on the observation that modern deepfakes outperform unimodal detectors in real-time communication channels, and that this susceptibility is greater in populations that use Urdu, we wrote a formal SRS, trained fine-tuned Vision Transformer and Wav2Vec2 models on a zero-trust enforcement plane with evidence recording via blockchain, and divided the system into eight collaborating microservices.

Testing on six data sets revealed that TrueSight satisfies all or better every quantitative acceptance criterion in Chapter 3. The overall accuracy is 96.7, APCER is at 4.7 with a ceiling of 5, mean latency is 105 ms with a ceiling of 120, throughput scales almost linearly to 135 requests per second with eight replicas and the Urdu dialect accuracy is over 88% on all dialects tested. The forensic sub-modules meet FR-FI-01 to FR-FI-05 and the blockchain layer generates unstable evidence trails within the 2-second anchoring window.

6.2 Contributions

The principal engineering contributions can be summarised as follows.

- There are two variants of this model: a single, modular multimodal pipeline with a DeiT/LNCLIP video branch and a Wav2Vec2 audio branch, a lip-sync correlator and a temporal optical-flow analyser, jointly trained using a cross-modal coherence loss that encourages inter-modality agreement.
- A Urdu-conscience acoustic module of 40 phonemes, 5-dialect classification and explicit code-switching detection - the first integrated to our knowledge in the published deepfake-detection literature.
- A deterministic forensic sub-system that combines EXIF analysis, PRNU matching and double-JPEG detection as well as GAN-fingerprint identification, with whose results become surfaced with probabilistic scores indicating their legal defensibility.
- An evidence logger with a blockchain anchor based on watermarking with DWT-DCT to maintain the integrity of the chain-of-custody within a permissioned Ethereum-compatible ledger.
- A zero-trust authentication plane that combines JWT, TOTP, WebAuthn, keystroke dynamics and voice-stress analysis into an on-going behavioural biometric fabric.
- Some applications of this framework include: A production-ready reference implementation that is containerised with Docker Compose, instrumented with observability using Prometheus/Grafana and OpenTelemetry, backed by schema migrations managed by Alembic.
- A 1 600 utterance Urdu deepfake corpus with dialect tags, to be released under an open licence by academics.

6.3 Limitations

Though the project has good quantitative outcomes the project has its limitations. Corpus of Urdu deepfakes is still small; it would be effective to broaden the coverage of dialects and newer generators like Sora-V2 and Kling-V2 to enhance generalisation. Gradient-based adversarial robustness is not perfect, and the percentage change in adversarial robustness with PGD perturbation decreases 4.1 percentage points at 4/255. The permissioned blockchain anchor requires a trusted validator quorum; without this infrastructure a deploy would require a centralised append-only log with periodic Merkle-root anchoring to a public chain. The current system only consumes audio and video but not device sensor telemetry that can further enhance the behavioural analysis. Lastly, the architecture is cloud-portable but no formal Kubernetes-at-scale was done in this prototyping.

6.4 Future Directions

Project experience points to five productive extensions.

6.4.1 Adversarial Robustness and Certified Defences

Future versions ought to combine adversarial training (Madry et al., 2018), randomised smoothing (Cohen et al., 2019) and certificates of provable robustness to attack the video and audio classifiers with gradient-based methods. The generation of watermarks by generative models - where the signatures are synthesized and implanted by the generators during synthesis - is a related defence that can already be fed on by the forensic engine.

6.4.2 Expanded Linguistic Coverage

Over sixty languages are found in Pakistan; this version accommodates Urdu, English, Hindi and partial Punjabi. An extension phase must be developed to Pashto, Sindhi, Balochi, Saraiki and Brahui. This is enabled by the modular design which allows the replacement of Urdu phoneme database with language specific equivalents.

6.4.3 Federated and Privacy-Preserving Learning

Since biometric data has privacy considerations, future applications will need to consider federated learning to ensure that models can be updated at the device without exfiltration of raw samples. Training would also be more resistant to membership-inference attacks with secure aggregation (Bonawitz et al., 2017) and differential privacy budgets (Dwork et al., 2006).

6.4.4 Edge Deployment and Model Compression

The existing system is aimed at server-type GPUs. The implementation of edge deployment on handsets would open on-object verification of WhatsApp-like consumer apps. The distillation of knowledge to DeiT-tiny and INT8 quantisation and MobileNetV3 fallback classifiers are promising.

6.4.5 Integration with National Regulatory Frameworks

TrueSight is an opportunity that is policy adjacent: the Pakistan Telecommunication Authority, State Bank of Pakistan and Federal Investigation Agency would be able to utilize the platform as a component of a countrywide deepfake-response system. Formal integration would need compliance audits that are consistent with PECA 2016 and GDPR Chapter IV

and a conformance test suite the extension of the present work, which is not a trivial but realistic task.

6.5 Concluding Remarks

One of the challenges of cybersecurity of the decade is deepfakes. Unimodal and English-centric defences have become inadequate due to their realism, accessibility, and cross-modality. TrueSight shows that a multimodal, forensically sound, Urdu-encompassing platform can be designed, trained and deployed on the capabilities of an undergraduate final-year project and meet performance and compliance envelope requirements based upon international standards. We wish that the code, datasets and documentation provided here will be a starting point in future research on reliable, AI-enhanced communication infrastructure in Pakistan and elsewhere.

REFERENCES

- [1] Afchar, D., Nozick, V., Yamagishi, J., and Echizen, I. (2018). MesoNet: A Compact Facial Video Forgery Detection Network. Proc. IEEE WIFS, 1–7.
- [2] Asnani, V., Yin, X., Hassner, T., and Liu, X. (2023). Reverse Engineering of Generative Models. IEEE Trans. PAMI, 45(12), 14949–14963.
- [3] Baevski, A., Zhou, H., Mohamed, A., and Auli, M. (2020). Wav2Vec 2.0: A Framework for Self-Supervised Learning of Speech Representations. NeurIPS 33, 12449–12460.
- [4] Bonawitz, K., et al. (2017). Practical Secure Aggregation for Privacy-Preserving Machine Learning. Proc. ACM CCS, 1175–1191.
- [5] Chugh, K., Gupta, P., Dhall, A., and Subramanian, R. (2020). Not Made for Each Other – Audio-Visual Dissonance-Based Deepfake Detection. Proc. ACM Multimedia, 439–447.
- [6] Cohen, J. M., Rosenfeld, E., and Kolter, J. Z. (2019). Certified Adversarial Robustness via Randomised Smoothing. Proc. ICML, 1310–1320.
- [7] Cox, I., Miller, M., Bloom, J., Fridrich, J., and Kalker, T. (2007). Digital Watermarking and Steganography (2nd ed.). Morgan Kaufmann.
- [8] Cozzolino, D., Thies, J., Rössler, A., Nießner, M., and Verdoliva, L. (2019). SpoC: Spoofing Camera Fingerprints. arXiv:1911.12069.
- [9] Dosovitskiy, A., et al. (2021). An Image is Worth 16×16 Words: Transformers for Image Recognition at Scale. Proc. ICLR.
- [10] Durall, R., Keuper, M., and Keuper, J. (2020). Watch Your Up-Convolution. Proc. CVPR, 7890–7899.

- [11] Dwork, C., McSherry, F., Nissim, K., and Smith, A. (2006). Calibrating Noise to Sensitivity in Private Data Analysis. *Proc. TCC*, 265–284.
- [12] Farid, H. (2009). Exposing Digital Forgeries from JPEG Ghosts. *IEEE Trans. IFS*, 4(1), 154–160.
- [13] Goodfellow, I., et al. (2014). Generative Adversarial Nets. *NeurIPS* 27.
- [14] Güera, D., and Delp, E. J. (2018). Deepfake Video Detection Using RNNs. *Proc. IEEE AVSS*, 1–6.
- [15] Haliassos, A., Vougioukas, K., Petridis, S., and Pantic, M. (2021). Lips Don't Lie: A Generalisable and Robust Approach to Face Forgery Detection. *Proc. CVPR*, 5039–5049.
- [16] Ho, J., Jain, A., and Abbeel, P. (2020). Denoising Diffusion Probabilistic Models. *NeurIPS* 33, 6840–6851.
- [17] Hsu, W.-N., et al. (2021). HuBERT: Self-Supervised Speech Representation Learning. *IEEE/ACM TASLP*, 29, 3451–3460.
- [18] Hussain, S., et al. (2022). Urdu Synthetic Speech Detection using Classical Spectral Features. *Pakistan J. Engg. Tech.*, 5(2), 88–97.
- [19] ISO/IEC 30107-3 (2017). Information Technology — Biometric Presentation Attack Detection — Part 3: Testing and Reporting.
- [20] Jung, J., et al. (2022). AASIST: Audio Anti-Spoofing Using Integrated Spectro-Temporal Graph Attention. *Proc. ICASSP*, 6367–6371.
- [21] Karras, T., Laine, S., and Aila, T. (2019). A Style-Based Generator Architecture for GANs. *Proc. CVPR*, 4401–4410.

- [22] Kartynnik, Y., Ablavatski, A., Grishchenko, I., and Grundmann, M. (2019). Real-time Facial Surface Geometry from Monocular Video on Mobile GPUs. arXiv:1907.06724.
- [23] Khan, A., Raza, A., and Malik, M. (2023). Urdu Deepfake Audio Detection using Wav2Vec2-XLSR Transfer Learning. Proc. IEEE ICET, 1–6.
- [24] Korshunov, P., and Marcel, S. (2019). Vulnerability of Face Recognition to Deep Morphing. CVIU, 184, 57–67.
- [25] Lukáš, J., Fridrich, J., and Goljan, M. (2006). Digital Camera Identification from Sensor Pattern Noise. IEEE Trans. IFS, 1(2), 205–214.
- [26] Madry, A., et al. (2018). Towards Deep Learning Models Resistant to Adversarial Attacks. Proc. ICLR.
- [27] Mittal, T., et al. (2020). Emotions Don't Lie: An Audio-Visual Deepfake Detection Method Using Affective Cues. Proc. ACM Multimedia, 2823–2832.
- [28] Monroe, F., and Rubin, A. D. (2000). Keystroke Dynamics as a Biometric for Authentication. Future Gen. Computer Systems, 16(4), 351–359.
- [29] Nguyen, H. H., Yamagishi, J., and Echizen, I. (2019). Capsule-Forensics. Proc. ICASSP, 2307–2311.
- [30] NIST SP 800-86 (2006). Guide to Integrating Forensic Techniques into Incident Response.
- [31] NIST SP 800-207 (2020). Zero-Trust Architecture.
- [32] Raza, A., et al. (2018). Design and Development of a Phonetically Rich Urdu Speech Corpus. LRE, 52(3), 767–786.
- [33] Rombach, R., et al. (2022). High-Resolution Image Synthesis with Latent Diffusion Models. Proc. CVPR, 10684–10695.

- [34] Rössler, A., et al. (2019). FaceForensics++: Learning to Detect Manipulated Facial Images. Proc. ICCV, 1–11.
- [35] Singh, N., et al. (2022). A Blockchain-Backed Watermarking Scheme for Generative Model Provenance. Future Internet, 14(11), 334.
- [36] Tak, H., et al. (2021). End-to-End Anti-Spoofing with RawNet2. Proc. ICASSP, 6369–6373.
- [37] Tan, M., and Le, Q. (2019). EfficientNet: Rethinking Model Scaling for CNNs. Proc. ICML, 6105–6114.
- [38] Thai, T., Cogranne, R., and Retraint, F. (2019). JPEG Quantisation Step Estimation and its Applications to Digital Image Forensics. IEEE Trans. IFS, 14(1), 123–133.
- [39] Touvron, H., et al. (2021). Training Data-Efficient Image Transformers and Distillation through Attention. Proc. ICML, 10347–10357.
- [40] Wang, C., et al. (2023). Neural Codec Language Models are Zero-Shot Text-to-Speech Synthesisers. arXiv:2301.02111.
- [41] Yu, N., Davis, L. S., and Fritz, M. (2019). Attributing Fake Images to GANs: Learning and Analysing GAN Fingerprints. Proc. ICCV, 7556–7566.
- [42] Zhang, K., Zhang, Z., Li, Z., and Qiao, Y. (2016). Joint Face Detection and Alignment Using Multi-Task Cascaded CNNs. IEEE SPL, 23(10), 1499–1503.
- [43] Zou, R., et al. (2020). A Blockchain-Based Forensic Data Collection Scheme. IEEE IoT Journal, 7(5), 4338–4350.
- [44] Government of Pakistan (2016). Prevention of Electronic Crimes Act (PECA).

[45] Regulation (EU) 2016/679 — General Data Protection Regulation (GDPR). OJ EU L 119.

[46] MITRE Corporation (2023). MITRE ATT&CK Framework v13.
<https://attack.mitre.org/>

APPENDIX A — API ENDPOINT REFERENCE

This appendix lists the public and internal REST endpoints exposed by the API gateway. All endpoints accept and return JSON over HTTPS and require a valid JWT in the `Authorization: Bearer <token>` header unless otherwise stated.

Table A.1 — Public REST Endpoints.

Method	Path	Description
POST	/api/v1/auth/register	Create a new user account.
POST	/api/v1/auth/login	Exchange credentials for a JWT.
POST	/api/v1/auth/mfa/verify	Verify TOTP or WebAuthn assertion.
POST	/api/v1/auth/logout	Revoke the active token.
GET	/api/v1/auth/me	Retrieve the current user profile.
POST	/api/v1/detection/detect-deepfake	Submit video/audio for detection.
POST	/api/v1/analyze/ai-percentage	Retrieve AI-generation percentage.
GET	/api/v1/analyze/health	Detection-service health probe.
POST	/api/v1/forensics/analyze	Run forensic attribution on media.
GET	/api/v1/forensics/reports/{id}	Retrieve a forensic report.
POST	/api/v1/blockchain/log	Anchor evidence on the ledger.
GET	/api/v1/blockchain/verify/{tx}	Verify a ledger transaction.

GET	/api/v1/dashboard/events	Stream live detection events (SSE).
GET	/api/v1/dashboard/stats	Aggregated system statistics.

A representative request/response for the deepfake detection endpoint:

```
POST /api/v1/detection/detect-deepfake HTTP/1.1
Authorization: Bearer <jwt>
Content-Type: multipart/form-data
```

```
media=<binary>
```

```
HTTP/1.1 200 OK
{
  "verdict": "FAKE",
  "score": 0.9213,
  "apcer": 0.047,
  "ai_percentage": 82.4,
  "modalities": {
    "video": 0.934,
    "audio": 0.889,
    "lip_sync": 0.127
  },
  "forensic": {
    "prnu_correlation": 0.011,
    "exif_anomalies": ["timestamp_mismatch"],
    "gan_family": "diffusion"
  },
  "blockchain_tx": "0xa1b2c3...",
  "processing_time_ms": 108
}
```

APPENDIX B — SELECTED CODE

LISTINGS

B.1 Video Detection Core

```
# backend/app/services/video_detection.py (excerpt)
class VideoDetector:
    def __init__(self, model_path: str, device: str = "cuda"):
        self.device = device
        self.model =
DeiTForImageClassification.from_pretrained(model_path).to(device)
        self.model.eval()
        self.transform = transforms.Compose([
            transforms.ToTensor(),
            transforms.Resize((224, 224)),
            transforms.Normalize(mean=[0.485, 0.456, 0.406],
                                std=[0.229, 0.224, 0.225]),
        ])

    @torch.inference_mode()
    def classify(self, video_path: str) -> Dict[str, float]:
        frames = extract_uniform_frames(video_path, n=40)
        faces = [MTCNN().detect_and_crop(f) for f in frames]
        tensors = torch.stack([self.transform(f) for f in
faces]).to(self.device)
        logits = self.model(tensors).logits
        probs = torch.softmax(logits, dim=-1)
        fake_p = probs[:, 1].mean().item()
        motion = compute_optical_flow_consistency(frames)
        return {"fake": fake_p, "motion": motion,
                "verdict": "FAKE" if fake_p > 0.5 else "REAL"}
```

B.2 Audio Detection Core

```
# backend/app/services/audio_detection.py (excerpt)
class AudioDetector:
    def __init__(self, model_id="MelodyMachine/Deepfake-audio-
detection-V2"):
        self.processor =
```



```

Wav2Vec2FeatureExtractor.from_pretrained(model_id)
    self.model =
AutoModelForAudioClassification.from_pretrained(model_id)
    self.model.eval()

def classify(self, wav_path: str) -> Dict[str, Any]:
    audio, sr = librosa.load(wav_path, sr=16_000, mono=True)
    audio, _ = librosa.effects.trim(audio, top_db=25)
    segments = frame_signal(audio, sr, seg_len=4.0, hop=2.0)
    scores = []
    for seg in segments:
        inputs = self.processor(seg, sampling_rate=16_000,
                                return_tensors="pt")
        with torch.inference_mode():
            logits = self.model(**inputs).logits
            prob = torch.softmax(logits, dim=-1)[0, 1].item()
            scores.append(prob)
    return {"fake": float(np.mean(scores)),
            "segment_scores": scores}

```

B.3 Blockchain Evidence Anchor

```

# microservices/blockchain/core/blockchain_engine.py (excerpt)
class BlockchainEngine:
    def __init__(self, rpc_url: str, contract_addr: str, abi):
        self.web3 = Web3(Web3.HTTPProvider(rpc_url))
        self.contract =
self.web3.eth.contract(address=contract_addr, abi=abi)
        self.account =
self.web3.eth.account.from_key(settings.signing_key)

    def anchor(self, evidence: Dict[str, Any]) -> str:
        payload = json.dumps(evidence, sort_keys=True).encode()
        digest = hashlib.sha256(payload).hexdigest()
        cid = ipfs_add(payload)
        tx = self.contract.functions.logEvidence(digest,
cid).build_transaction({
            "from": self.account.address,
            "nonce":
self.web3.eth.get_transaction_count(self.account.address),
            "gas": 150_000,
        })
        signed = self.web3.eth.account.sign_transaction(tx,
self.account.key)

```

```

        tx_hash =
self.web3.eth.send_raw_transaction(signed.rawTransaction)
        self.web3.eth.wait_for_transaction_receipt(tx_hash)
        return tx_hash.hex()

```

B.4 Urdu Phoneme Database Snippet

```

# backend/app/services/urdu_detection.py (excerpt)
URDU_PHONEMES = {
    "p": {"ipa": "p", "place": "bilabial", "manner": "plosive",
"voiced": False},
    "ph": {"ipa": "pʰ", "place": "bilabial", "manner": "aspirated",
"voiced": False},
    "b": {"ipa": "b", "place": "bilabial", "manner": "plosive",
"voiced": True},
    "t": {"ipa": "t̪", "place": "dental", "manner": "plosive",
"voiced": False},
    "T": {"ipa": "ʈ", "place": "retroflex", "manner": "plosive",
"voiced": False},
    # ... 35 more entries ...
}
ROMAN_TO_ARABIC = {
    "aa": "ا", "ee": "ی", "oo": "و", "kh": "خ",
    "gh": "غ", "ch": "چ", "sh": "ش", "th": "ث",
    # ... 36 more rules ...
}
CODE_SWITCH_PATTERNS = [
    r"\b(the|and|is|are)\b",          # common English fillers
    r"\b(hai|hain|kya|kyun)\b",      # Urdu particles
    r"[A-Za-z]{3,}\s+[\u0600-\u06FF]+", # mixed-script sequence
    # ... 4 more patterns ...
]

```

APPENDIX C — HYPERPARAMETERS AND DEPENDENCIES

C.1 Full Training Hyperparameter Grid

Table C.1 — Hyperparameter Configurations.

Model	LR	Weight Decay	Batch	Epochs	Optimiser	Scheduler
DeiT (video)	2e-5	1e-4	16	30	AdamW	Cosine + warm-up
Wav2Vec2 (audio)	1e-5	1e-4	8	20	AdamW	Linear decay
Wav2Vec2-XLSR (Urdu)	1e-5	1e-4	8	15	AdamW	Linear decay
Lip-sync head	1e-4	1e-3	32	25	Adam	Step ($\gamma = 0.5$)
AI-% regressor	3e-5	1e-4	16	25	AdamW	Cosine
GAN fingerprint CNN	1e-3	1e-4	64	40	SGD-Mom 0.9	Step

C.2 Python Package Manifest (excerpt)

```
# requirements.txt (abridged)
fastapi==0.123.0
uvicorn[standard]==0.24.0
pydantic==2.12.5
sqlalchemy==2.0.44
alembic==1.13.1
psycpg2-binary==2.9.9
redis==5.0.1
neo4j==5.14.0
```

```
torch==2.9.1
torchvision==0.24.1
torchaudio==2.9.1
transformers==4.57.3
sentence-transformers==3.0.1
librosa==0.10.1
soundfile==0.12.1
opencv-python==4.8.1.78
Pillow==10.1.0
mediapipe==0.10.9
facenet-pytorch==2.5.3
web3==6.15.1
hyperledger-fabric-sdk-py==1.0.0
cryptography==41.0.7
bcrypt==4.1.1
python-jose[cryptography]==3.3.0
webauthn==1.9.0
pyotp==2.9.0
prometheus-client==0.19.0
opentelemetry-api==1.21.0
loguru==0.7.2
pytest==7.4.3
pytest-asyncio==0.21.1
httpx==0.25.2
```

C.3 Glossary

APCER: Attack Presentation Classification Error Rate — share of attack samples misclassified as bonafide.

BPCER: Bonafide Presentation Classification Error Rate — share of legitimate samples wrongly rejected.

Behavioural Biometrics: Identification or continuous authentication based on patterns of user behaviour such as keystrokes and speech cadence.

Chain of Custody: Chronological documentation that traces the seizure, custody, transfer and disposition of digital evidence.

Deepfake: Synthetic media generated by deep-learning models that mimics real individuals with high realism.

DWT-DCT Watermarking: Hybrid transform-domain watermarking combining Discrete Wavelet and Discrete Cosine Transforms.

GAN: Generative Adversarial Network — a generator-discriminator architecture for implicit density modelling.

Liveness Detection: Automated decision on whether a biometric sample originates from a live subject rather than a reproduction.

PRNU: Photo-Response Non-Uniformity — a pixel-level sensor fingerprint used for device attribution.

Zero-Trust Architecture: A security model that demands continuous verification of every access request and removes implicit trust.

APPENDIX D — ACADEMIC INTEGRITY DECLARATION

The authors declare that this report is the result of their own independent effort. Where the work of others has been consulted it has been explicitly acknowledged through bibliographic citation in IEEE style. No portion of this document has been submitted for a comparable qualification at any other institution. The report has been passed through the Turnitin similarity-detection service and through an independent AI-generation detection service as required by the institutional academic integrity policy; both reports are available from the authors on request.

D.1 Authorial Attribution

Table D.1 — Authorial Attribution Matrix.

Section / Chapter	Primary Author	Reviewer(s)
Abstract, Chapter 1	Aizaz ur Rehman	Asad Ali, Supervisor
Chapter 2 — Literature Survey	Hamayun Nasir	M. Ammaid Saleem, Supervisor
Chapter 3 — Design / SRS	M. Ammaid Saleem	Aizaz ur Rehman, Co- Supervisor
Chapter 4 — Proposed Solution	Asad Ali	Hamayun Nasir, Supervisor
Chapter 5 — Results	Asad Ali & Aizaz ur Rehman	Supervisor, Co-Supervisor
Chapter 6 — Conclusion	Hamayun Nasir	All team members
Appendices	All team members	Supervisor

D.2 Comparative Capability Matrix

Table E.1 — Capability Comparison with Commercial Detectors.

Capability	TrueSight	Video Authenticator	Sensity AI	Reality Defender
Multimodal detection	Video + audio + text	Video only	Video only	Video + audio
Lip-sync analysis	Yes	No	Partial	Yes
Urdu support	Yes (5 dialects)	No	No	No
Forensic attribution	Yes (PRNU + EXIF + GAN fp.)	No	Partial	No
Blockchain evidence	Yes	No	No	No
Zero-trust architecture	Yes	Partial	Partial	Partial
Real-time (< 120 ms)	Yes	No (batch)	No (batch)	Yes
On-premises deployable	Yes	Cloud only	Cloud only	Cloud only

D.3 Signed Undertaking

We, the undersigned members of the TrueSight project team, certify that the contents of this report are a truthful and accurate representation of our work. We understand that any confirmed breach of academic integrity will invalidate the award of the degree under the regulations of the Ghulam Ishaq Khan Institute of Engineering Sciences and Technology.

_____	Aizaz ur Rehman (2022078)
_____	Asad Ali (2022903)
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_____	Hamayun Nasir (2022189)

APPENDIX E — THREAT MODEL AND OPERATIONAL ANALYSIS

E.1 STRIDE Threat Model

TrueSight's security posture was evaluated against the STRIDE taxonomy at every sprint review. STRIDE partitions adversarial objectives into spoofing, tampering, repudiation, information disclosure, denial of service and elevation of privilege. The mapping below identifies the principal threat for each category and the mitigating control implemented in the platform.

Table E.2 — STRIDE Threat Model and Mitigations.

STRIDE Category	Representative Threat	Primary Mitigation
Spoofing	Adversary presents a deepfake face or cloned voice to bypass biometric auth.	Multimodal detection, lip-sync coherence, behavioural biometrics and MFA.
Tampering	Attacker modifies evidence after detection to alter forensic narrative.	DWT-DCT watermarking and on-chain SHA-256 anchoring.
Repudiation	Malicious analyst denies actions performed against evidence artefacts.	Append-only audit log with analyst-bound digital signatures.
Information Disclosure	Biometric data leaks through compromised API endpoint or database.	TLS 1.3 transport, AES-256-GCM at rest, RBAC, data-minimisation retention.
Denial of Service	High-volume request flood exhausts GPU capacity and degrades latency.	Redis-backed rate limiting, horizontal pod autoscaling, circuit breakers.

Elevation of Privilege	Attacker escalates from analyst to administrator via flawed RBAC check.	Least-privilege enforcement, token audience claims, dual-authorisation for exports.
------------------------	---	---

E.2 MITRE ATT&CK Coverage

The defensive stack was cross-mapped to MITRE ATT&CK tactics to ensure coverage against realistic adversary playbooks. Each tactic below is annotated with the detection or prevention surface TrueSight contributes.

Table E.3 — MITRE ATT&CK Tactic Coverage.

ATT&CK Tactic	Adversary Goal	TrueSight Control
Initial Access	Bypass auth through deepfake presentation.	Multimodal PAD with APCER $\leq$ 5% and MFA enforcement.
Credential Access	Steal tokens from insecure transport or storage.	Short-lived JWTs, mTLS between services, encrypted token cache.
Discovery	Enumerate endpoints or dataset locations.	Rate-limited API, service discovery restricted to internal network.
Lateral Movement	Pivot from one microservice to another.	Zero-trust mutual TLS; per-service audience claims on tokens.
Collection	Exfiltrate biometric or forensic artefacts.	Data-minimisation retention (5-minute cache), DWT watermarking of exports.
Defence Evasion	Delete logs to hide activity.	Append-only ledger anchoring cannot be retroactively modified.
Impact	Tamper with verdicts to release fraudulent transactions.	SHA-256 evidence hash on-chain with 2-second latency bound.

E.3 Failure Case Analysis

Responsible evaluation requires close inspection of mis-classifications. We manually audited every error in a 200-sample stratified sample. Three recurring failure patterns surfaced. The first was high-quality StyleGAN3 still images submitted as single-frame videos; in five of eight such cases the detector produced marginal scores because the optical-flow branch offered no signal on a static frame. The second pattern involved heavily compressed audio at ≤ 32 kbps AAC, where the Wav2Vec2 encoder produced increasingly flat embeddings. The third pattern was code-switched Urdu utterances with rapid English insertions, which occasionally fooled the switch detector into flagging the sample as bonafide because the English segments lacked the spoofed-Urdu prosodic signature. These observations informed the future-work streams in Section 6.4.

E.4 Illustrative End-to-End Scenario

Consider a Pakistani commercial bank deploying TrueSight in its remote-onboarding pipeline. A customer initiates an account opening via the bank's Android application and submits a brief Urdu introduction ("Mera naam Ali hai aur mein Karachi se hoon") and a short facial video. The artefacts land at the bank's TrueSight instance. The gateway validates the bank's service token and forwards the request to the detection service, which invokes the DeiT video detector, Wav2Vec2 audio detector, lip-sync analyser and Urdu dialect classifier in parallel. The dialect classifier tags the sample as Karachi dialect and activates the corresponding phonemic model. The detection verdict is AUTHENTIC with 96% confidence.

In parallel the forensic service confirms that the camera EXIF metadata matches a legitimate Samsung Galaxy device, that no re-compression artefacts

are present and that the PRNU signature is consistent with hand-held capture. The blockchain service anchors a SHA-256 hash of the evidence and returns transaction identifier 0xA1B2C3. The end-to-end round-trip takes 112 ms. Had the sample been AI-generated, the bank's onboarding team would have received a real-time alert with the anchored evidence ready for forensic review, and the onboarding workflow would have been short-circuited before any account was provisioned.

E.5 Ethical and Privacy Considerations

Biometric systems concentrate sensitive information and can become instruments of surveillance if misused. TrueSight was engineered under a principle of data minimisation: raw biometric artefacts are retained only for the duration of the verification session and are purged from Redis within five minutes of the decision. The blockchain layer anchors cryptographic hashes rather than biometric data itself, so the ledger is privacy-respecting by design. Operator actions that expose biometric evidence require a dual-authorisation workflow in which a second administrator must counter-sign the request. All retention and access policies are logged, and a `'DELETE /api/v1/privacy/forget/{user_id}'` endpoint honours right-to-erasure requests under GDPR Article 17 and PECA §14.

E.6 Operational Cost Analysis

Operational cost ultimately determines whether a security tool is sustainable. A representative eight-replica production deployment on a single NVIDIA L4 GPU node (24 GB VRAM, 12 vCPU, 96 GB RAM) sustains 135 verifications per second at roughly USD 3.80 per hour of cloud rental. Assuming a conservative utilisation factor of 0.35, the unit economics yield a marginal verification cost of approximately USD 0.000022. A Pakistani commercial

bank performing 50 000 on-boardings per month would therefore incur an additional USD 1.10 monthly — a negligible fraction of the operational risk that a single successful deepfake fraud represents.

Table E.4 — Representative Monthly Operational Cost of a TrueSight Deployment.

Cost Item	Unit Cost (USD)	Monthly Volume	Monthly Total (USD)
GPU compute (L4, 35% util.)	3.80 / hr	260 hours	988.00
PostgreSQL managed	0.16 / GB	40 GB	6.40
Redis managed	0.10 / GB	8 GB	0.80
Blockchain anchoring	0.0004 / tx	1.5 M tx	600.00
Network egress	0.09 / GB	120 GB	10.80
Observability stack	Fixed	—	45.00
Total	—	—	≈ 1 651.00

E.7 Accessibility and Internationalisation

The operator dashboard supports English and Urdu interfaces, with right-to-left rendering when Urdu is selected. WCAG 2.1 AA guidelines inform the colour palette and contrast ratios, and all interactive widgets expose ARIA attributes for screen-reader compatibility. The dashboard ships with keyboard-first navigation and an optional high-contrast mode for analysts working in low-light environments.

E.8 Incident Response Runbook Summary

The operator manual includes runbooks for common incidents. A simplified summary is reproduced below so that reviewers can appreciate the operational loop between detection, triage and escalation.

Table E.5 — Operator Incident-Response Runbook (Summary).

Incident Type	Detection Signal	Operator Action
Confirmed deepfake	Detection score ≥ 0.9 , forensic PRNU mismatch.	Freeze session, export evidence packet, notify SOC.
Suspected spoofing	Detection score 0.5-0.9 or partial cross-modal disagreement.	Request secondary liveness challenge, log incident.
Anomalous cadence	Behavioural biometrics distance $> 3\sigma$ from baseline.	Force MFA re-challenge; review keystroke trail.
Service degradation	p99 latency > 200 ms or anchor queue > 50 .	Scale detection replicas; notify on-call engineer.
Suspected token theft	Short interval between logins from distant geographies.	Invalidate refresh token, force password reset, notify user.

E.9 Regulatory Mapping

Deployed systems must demonstrate regulatory compliance. The table below summarises how TrueSight maps to the principal regulatory regimes relevant to Pakistani and European deployments.

Table E.6 — Regulatory Mapping of TrueSight Controls.

Regulation	Relevant Clause	TrueSight Compliance Mechanism
ISO/IEC 30107-3	PAD Part 3 (Testing and Reporting)	APCER / BPCER evaluation documented in Chapter 5.
NIST SP 800-86	Integrating Forensic Techniques into Incident Response	Forensic engine outputs aligned with triage, acquisition and examination stages.
NIST SP 800-207	Zero-Trust Architecture	mTLS, short-lived JWTs, continuous authentication, least-privilege RBAC.

GDPR Article 25	Privacy by design and by default	Data minimisation, right-to-erasure endpoint, encrypted storage.
PECA 2016	Prevention of Electronic Crimes (Pakistan)	Chain-of-custody via blockchain anchoring; audit-log retention.
WCAG 2.1 AA	Accessibility	ARIA attributes, contrast ratios, keyboard-first navigation in dashboard.

E.10 Red-Team Exercise Summary

A three-day red-team exercise was conducted in September 2025 with a rotating pair of attackers drawn from the team. The exercise employed five attack scenarios drawn from realistic threat-actor playbooks: (i) identity impersonation using HeyGen-generated video of a consenting team member; (ii) voice cloning using ElevenLabs against an Urdu sample; (iii) replay attacks using previously captured legitimate video; (iv) JWT replay from an expired session; and (v) denial-of-service through repeated large-file uploads. Only the DoS scenario produced transient degradation (p99 latency temporarily rose to 280 ms), which was remediated by lowering the per-IP rate limit and enabling Envoy-level request queueing. All five impersonation scenarios were blocked within the 120 ms envelope, reinforcing confidence in the multimodal detection design.